

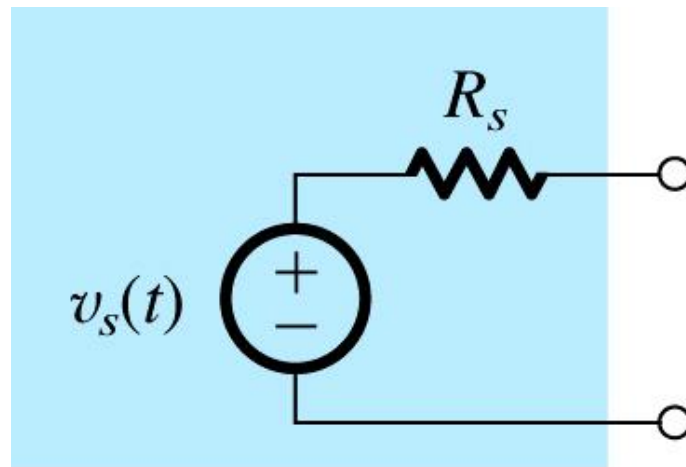
Fundamentals of Analogue Circuits

REVIEW

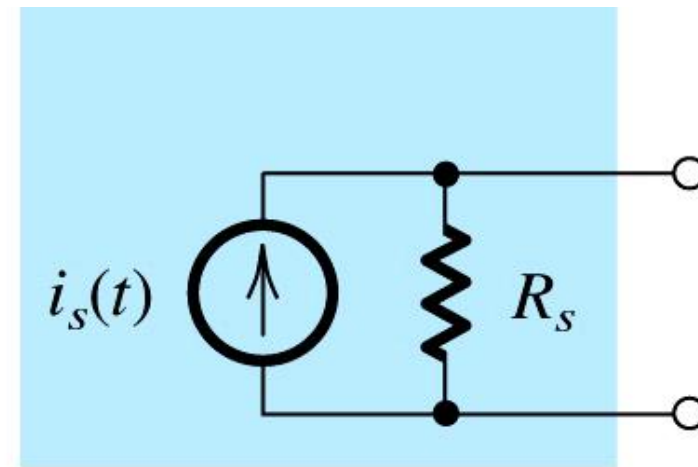
Chapter #1: Signals and Amplifiers

Common laws:

KCL, KVL, Superposition, Thevenin's theorem, Norton's theorem



(a)

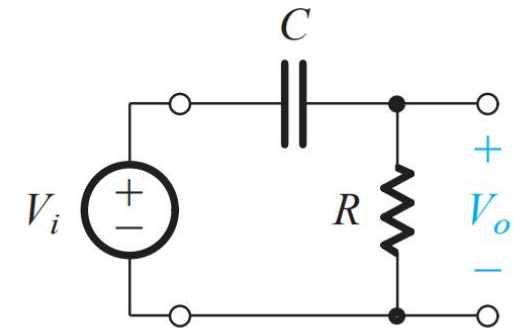
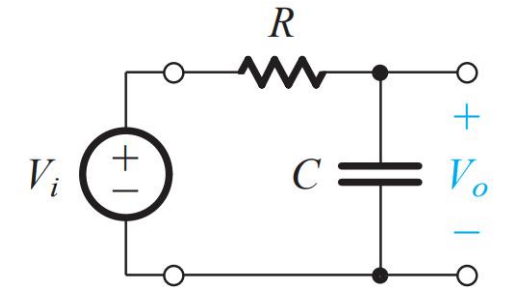


(b)

Figure 1.1: Two alternative representations of a signal source:
(a) the Thévenin form; (b) the Norton form.

Table 1.2 Frequency Response of STC Networks

	Low-Pass (LP)	High-Pass (HP)
Transfer Function $T(s)$	$\frac{K}{1 + (s/\omega_0)}$	$\frac{Ks}{s + \omega_0}$
Transfer Function (for physical frequencies) $T(j\omega)$	$\frac{K}{1 + j(\omega/\omega_0)}$	$\frac{K}{1 - j(\omega_0/\omega)}$
Magnitude Response $ T(j\omega) $	$\frac{ K }{\sqrt{1 + (\omega/\omega_0)^2}}$	$\frac{ K }{\sqrt{1 + (\omega_0/\omega)^2}}$
Phase Response $\angle T(j\omega)$	$-\tan^{-1}(\omega/\omega_0)$	$\tan^{-1}(\omega_0/\omega)$
Transmission at $\omega = 0$ (dc)	K	0
Transmission at $\omega = \infty$	0	K
3-dB Frequency	$\omega_0 = 1/\tau$; $\tau \equiv$ time constant $\tau = CR$ or L/R	
Bode Plots	in Fig. 1.23	in Fig. 1.24



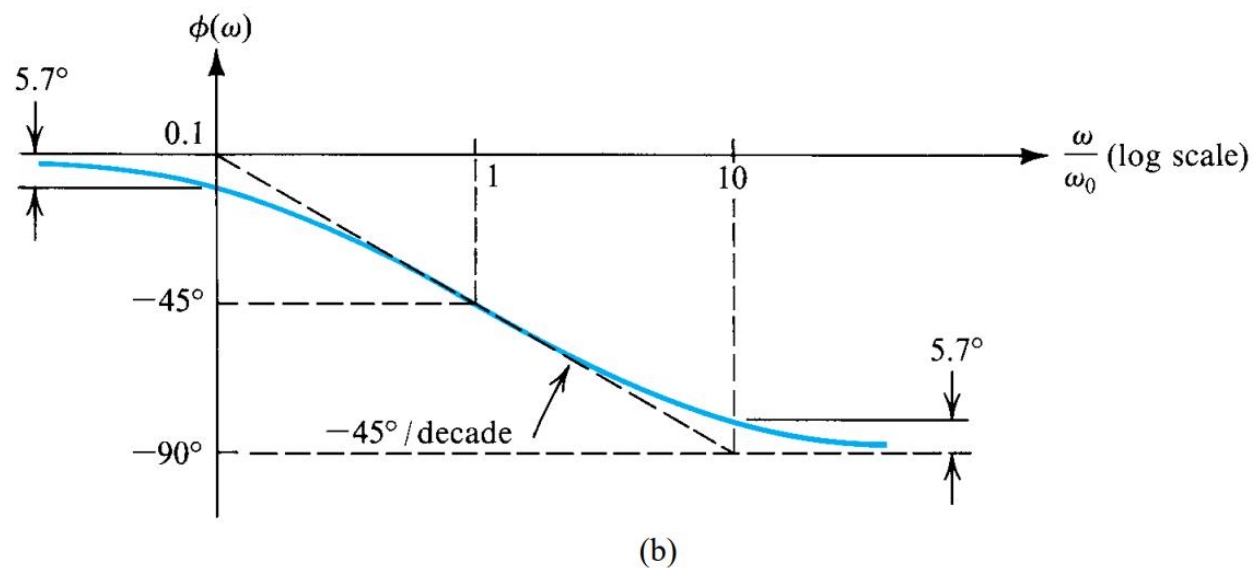
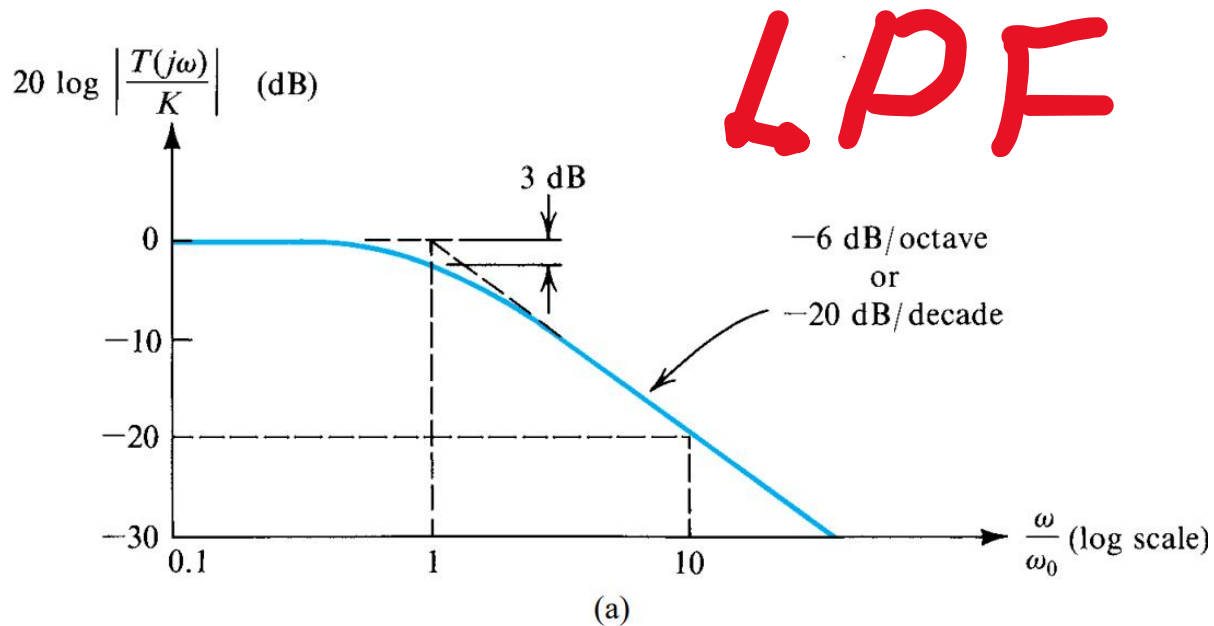


Figure 1.23 (a) Magnitude and (b) phase response of STC networks of the low-pass type.

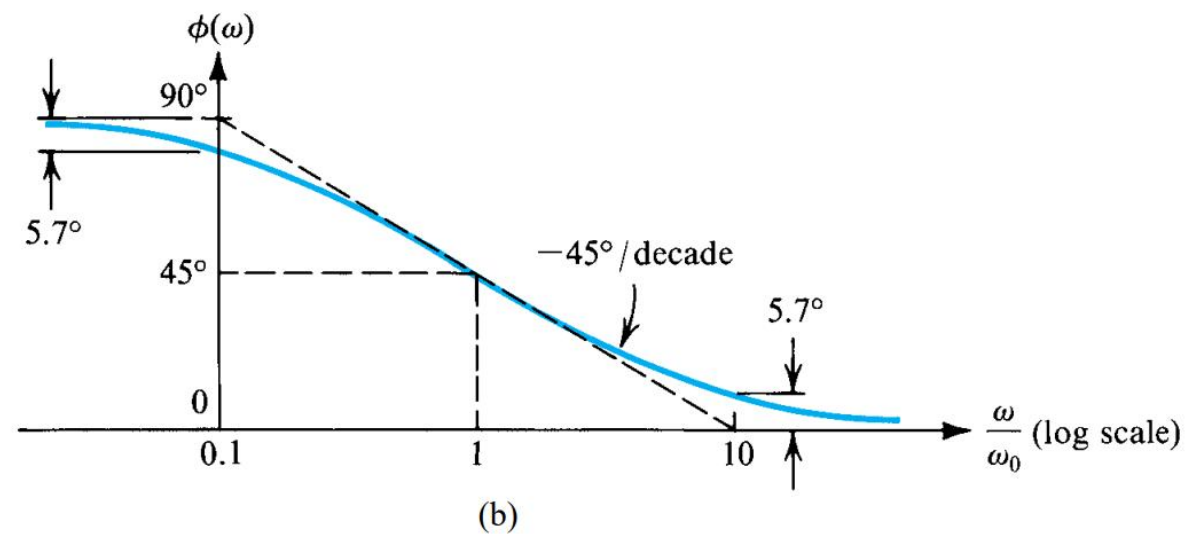
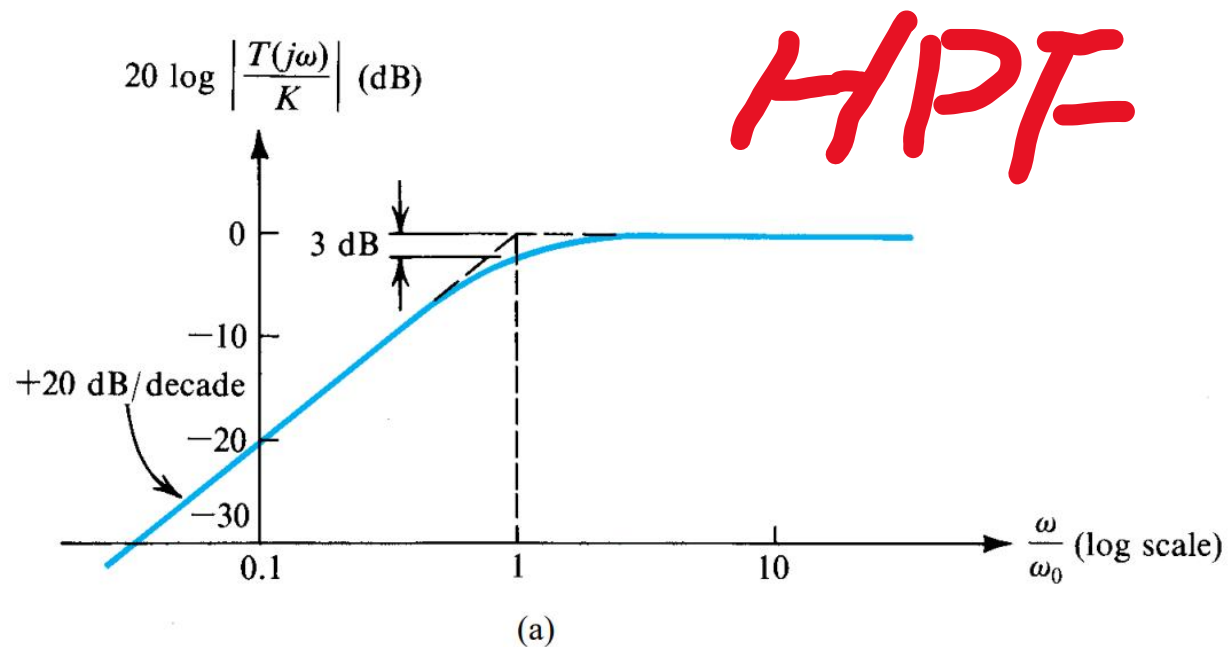
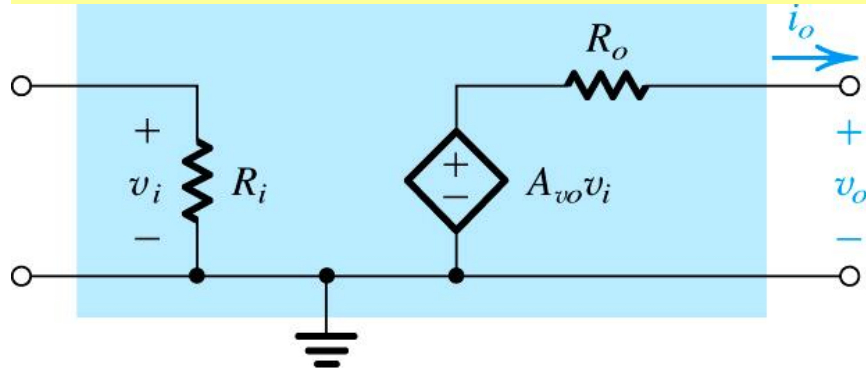


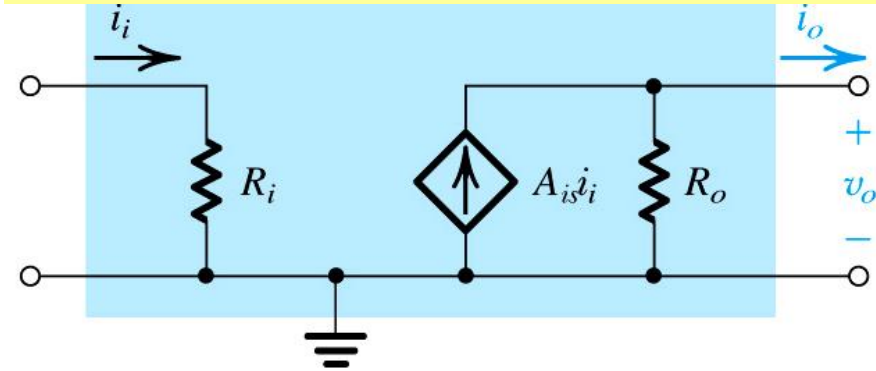
Figure 1.24 (a) Magnitude and (b) phase response of STC networks of the high-pass type.

1.5.3. Other Amplifier Types

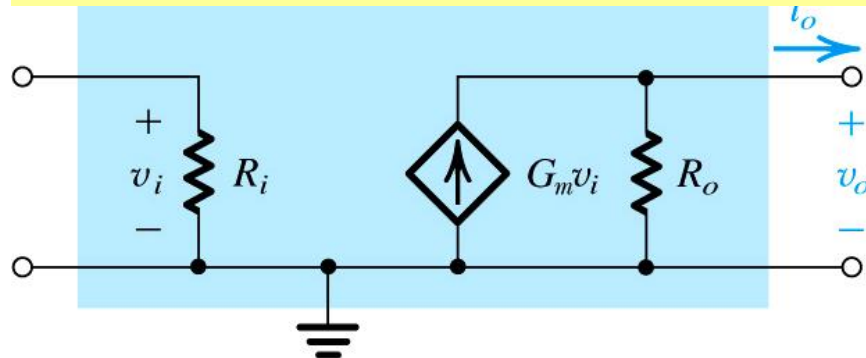
voltage amplifier



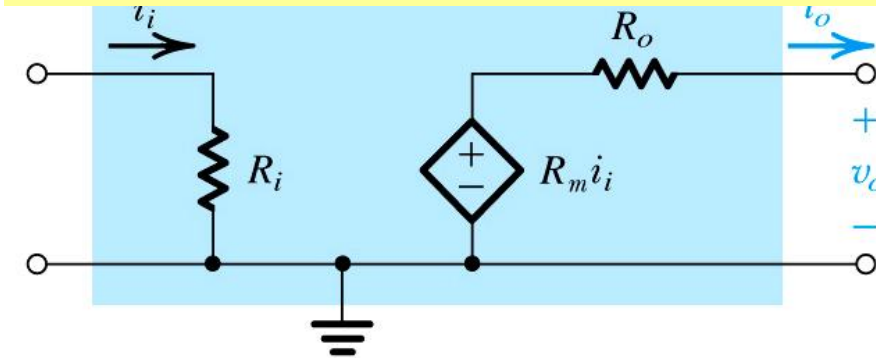
current amplifier



transconductance amp.

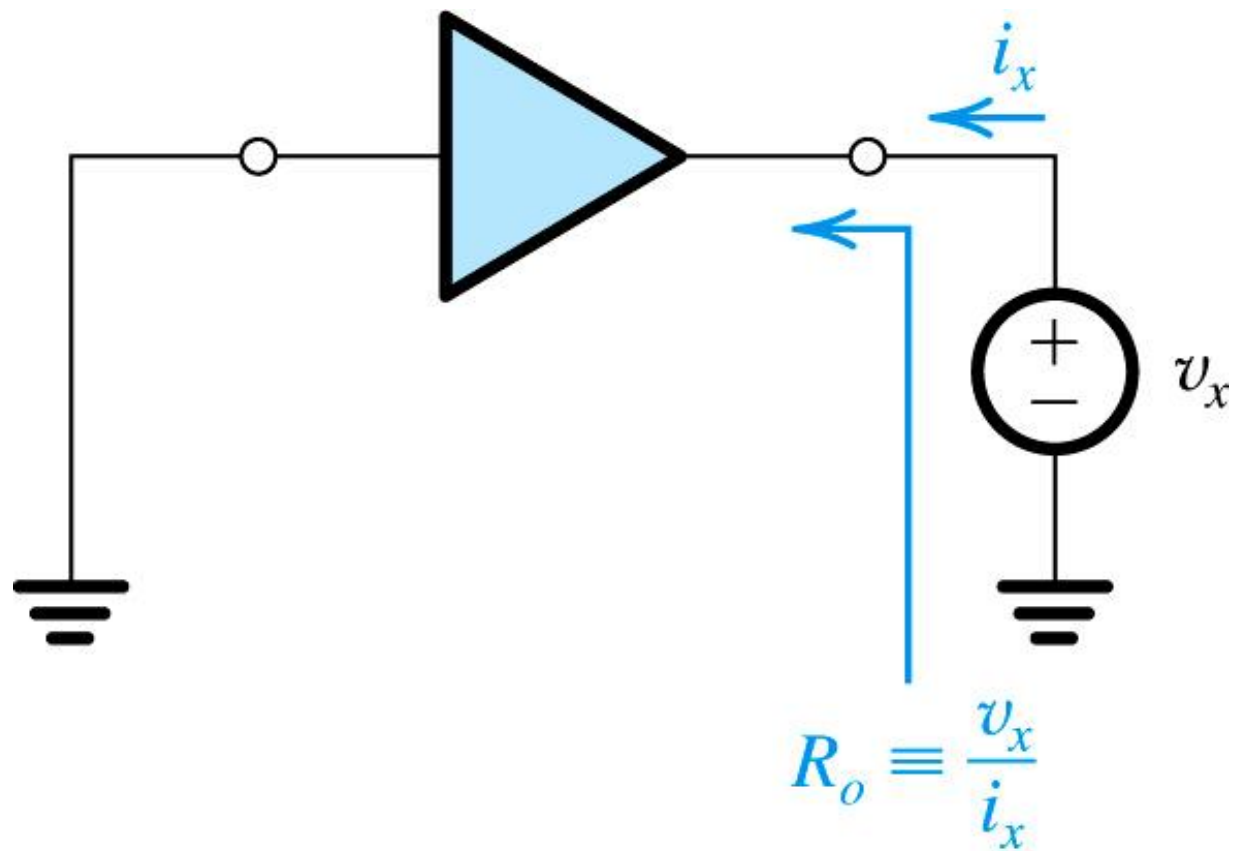


transresistance amp.

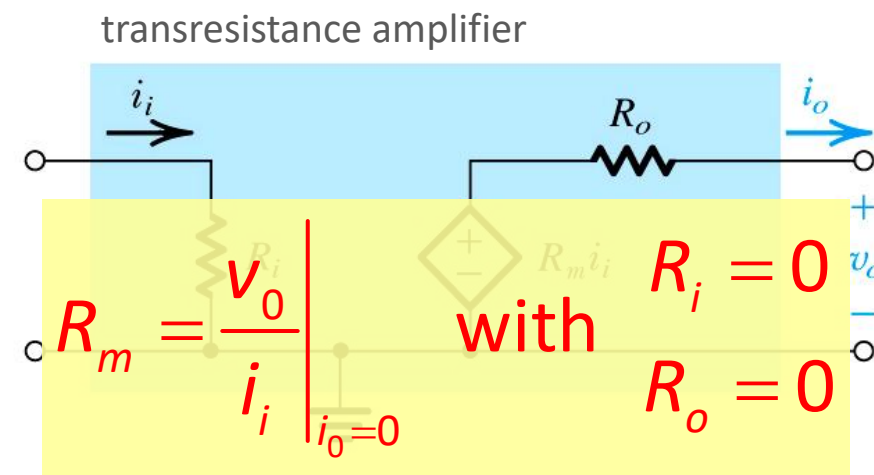
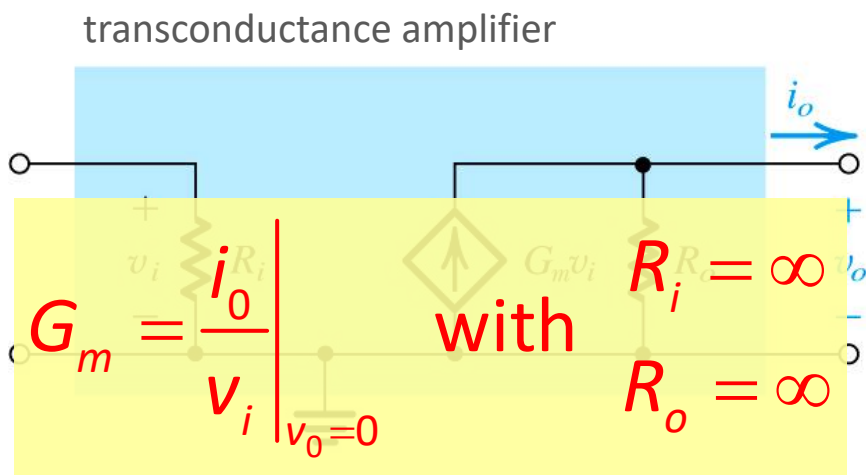
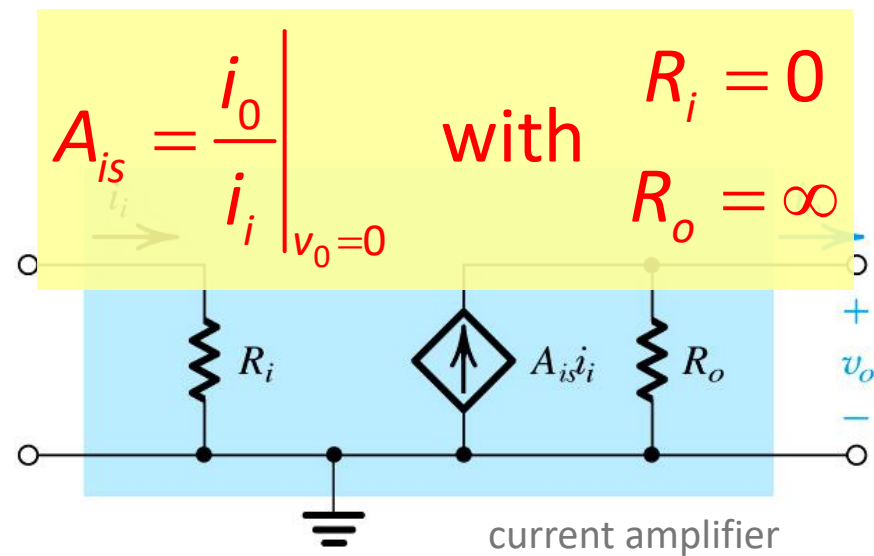
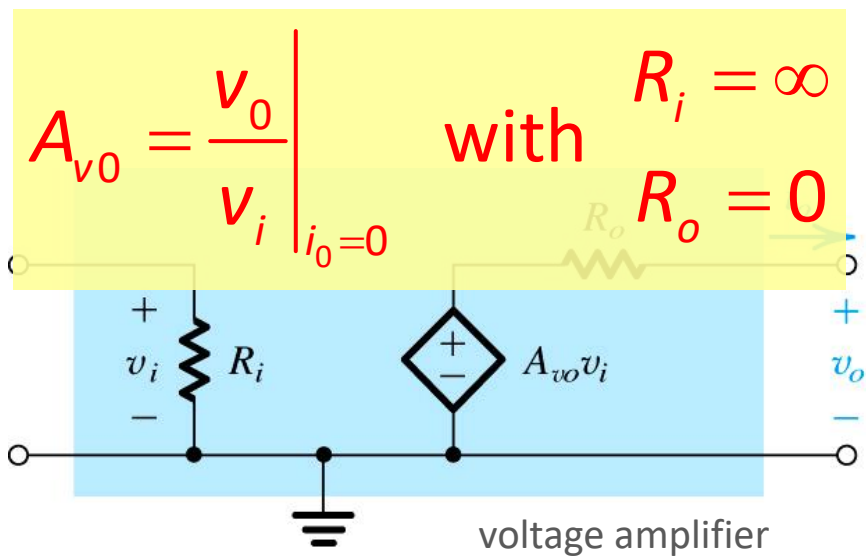


Determining R_i and R_o

- $R_i = v_i / i_i$



1.5.3. Other Amplifier Types



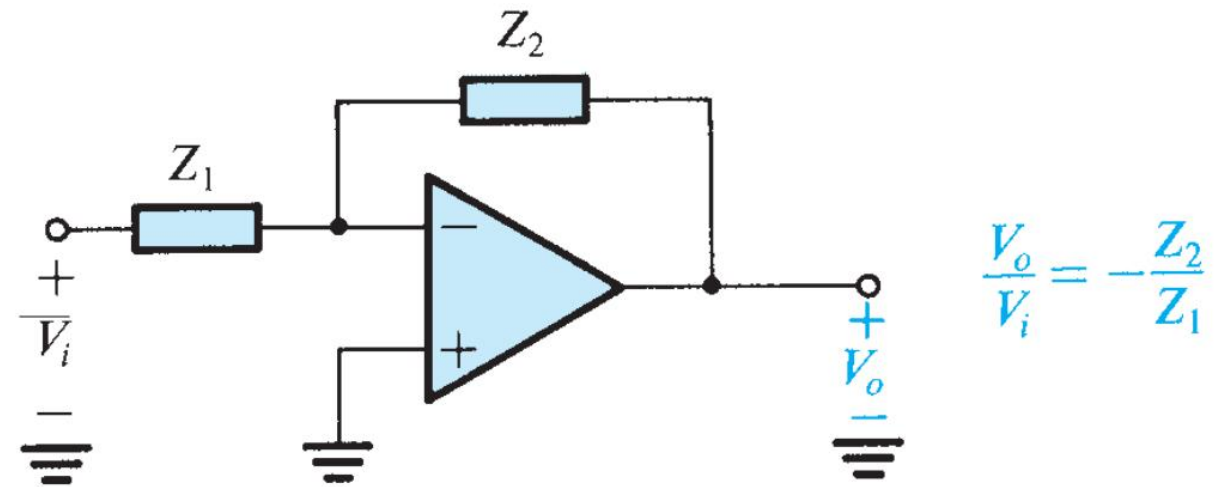
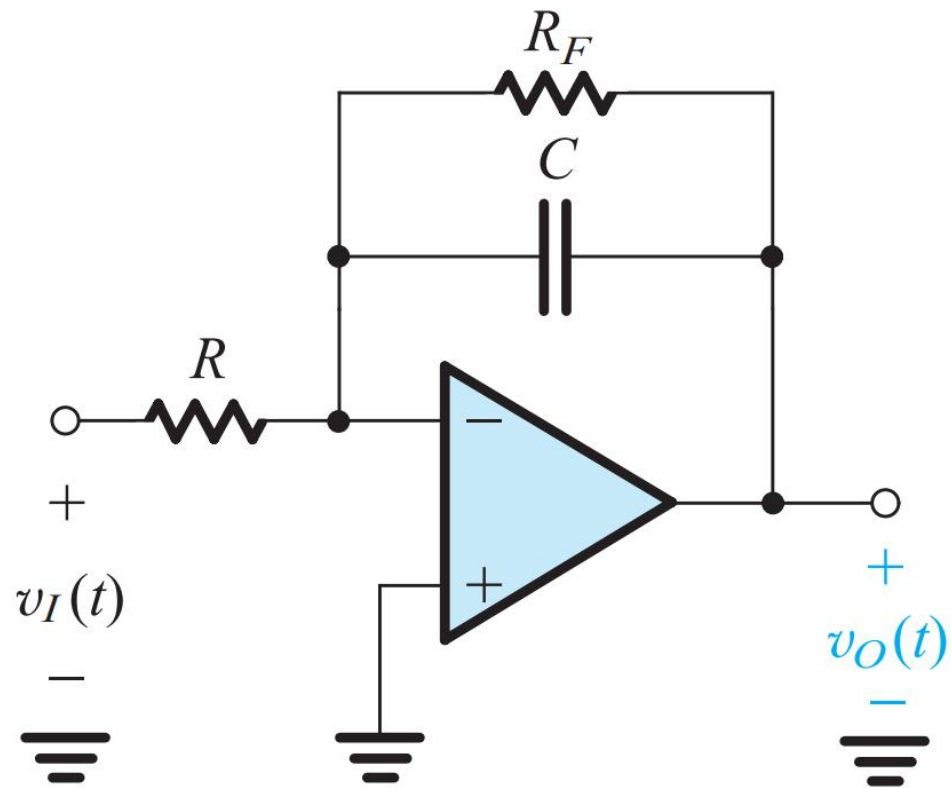


Figure 2.25 The Miller integrator with a large resistance R_F connected in parallel with C in order to provide negative feedback and hence finite gain at dc.

Example 2.4

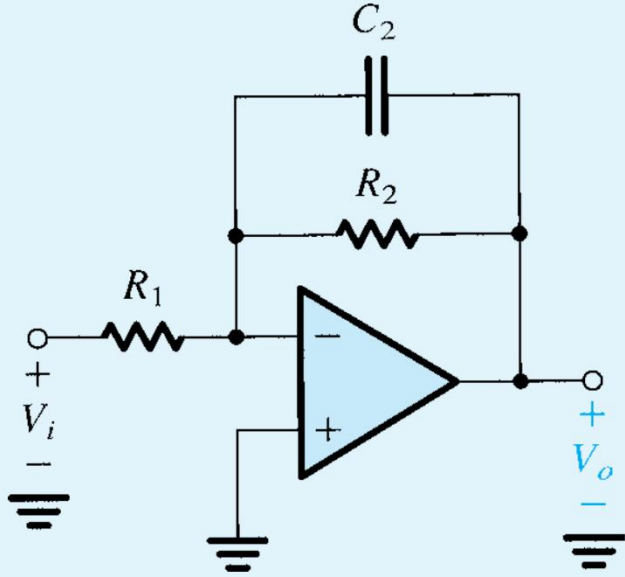


Figure 2.23 Circuit for Example 2.4.

$$Z_1 = R_1$$

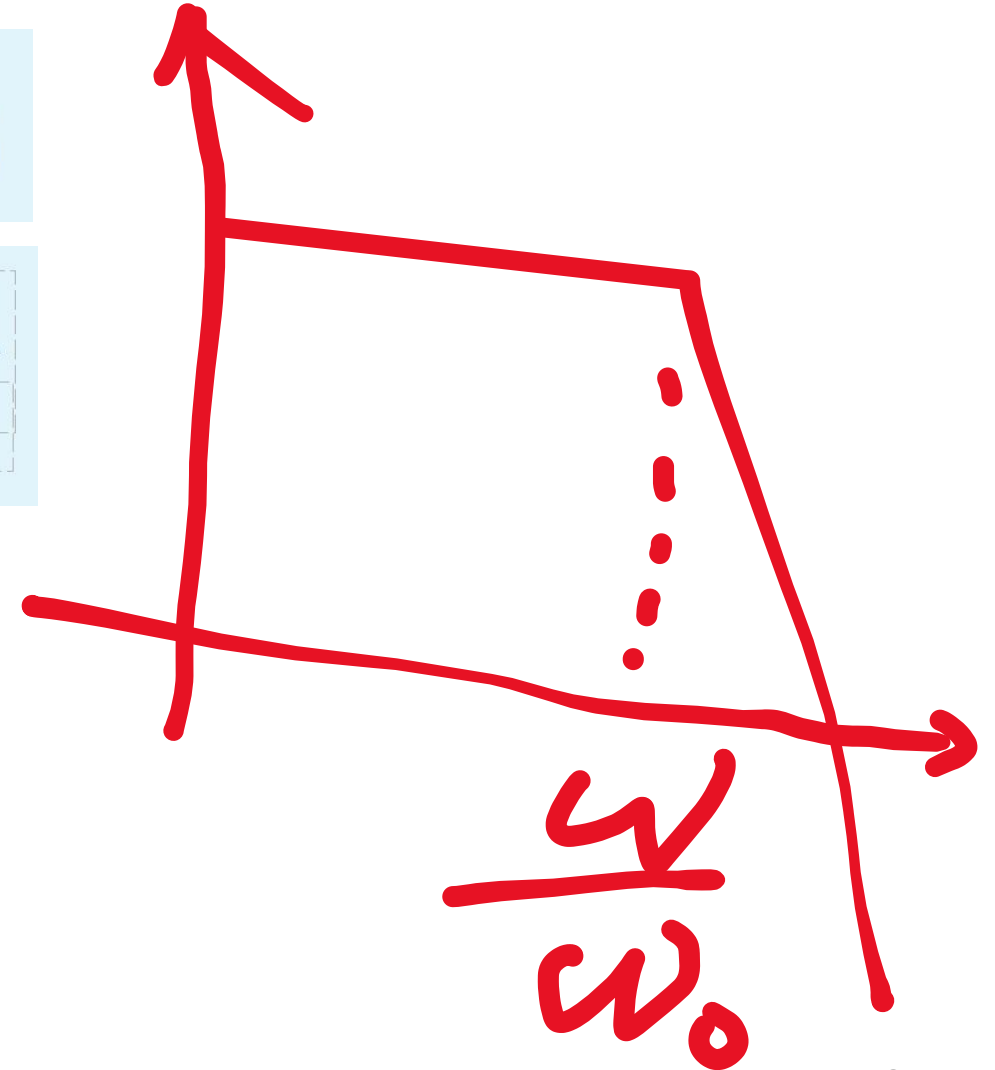
$$Z_2 = R_2 \parallel (1/sC_2)$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1}{Z_1(s)Y_2(s)}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1}{\frac{R_1}{R_2} + sC_2R_1}$$

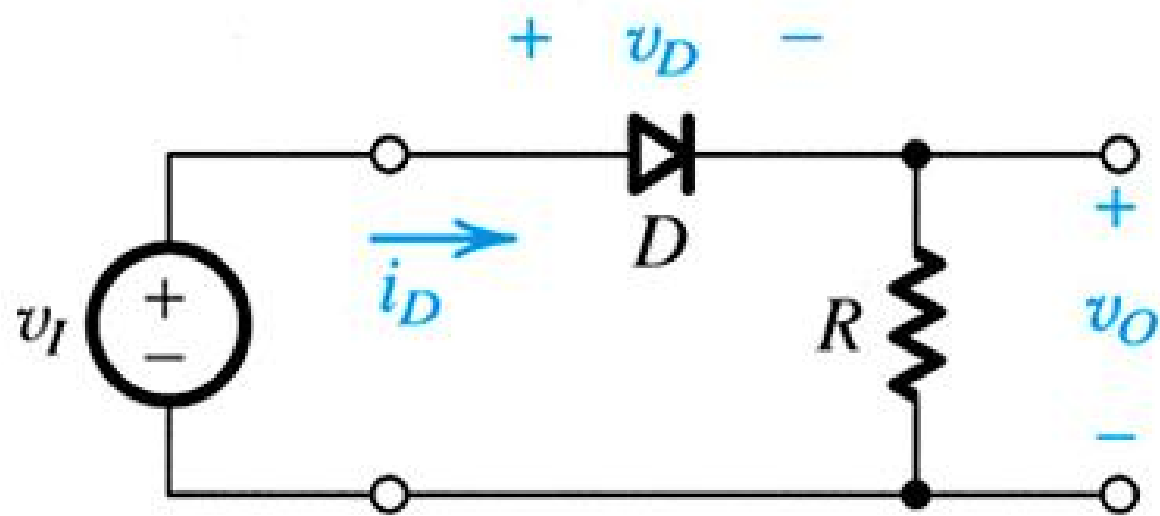
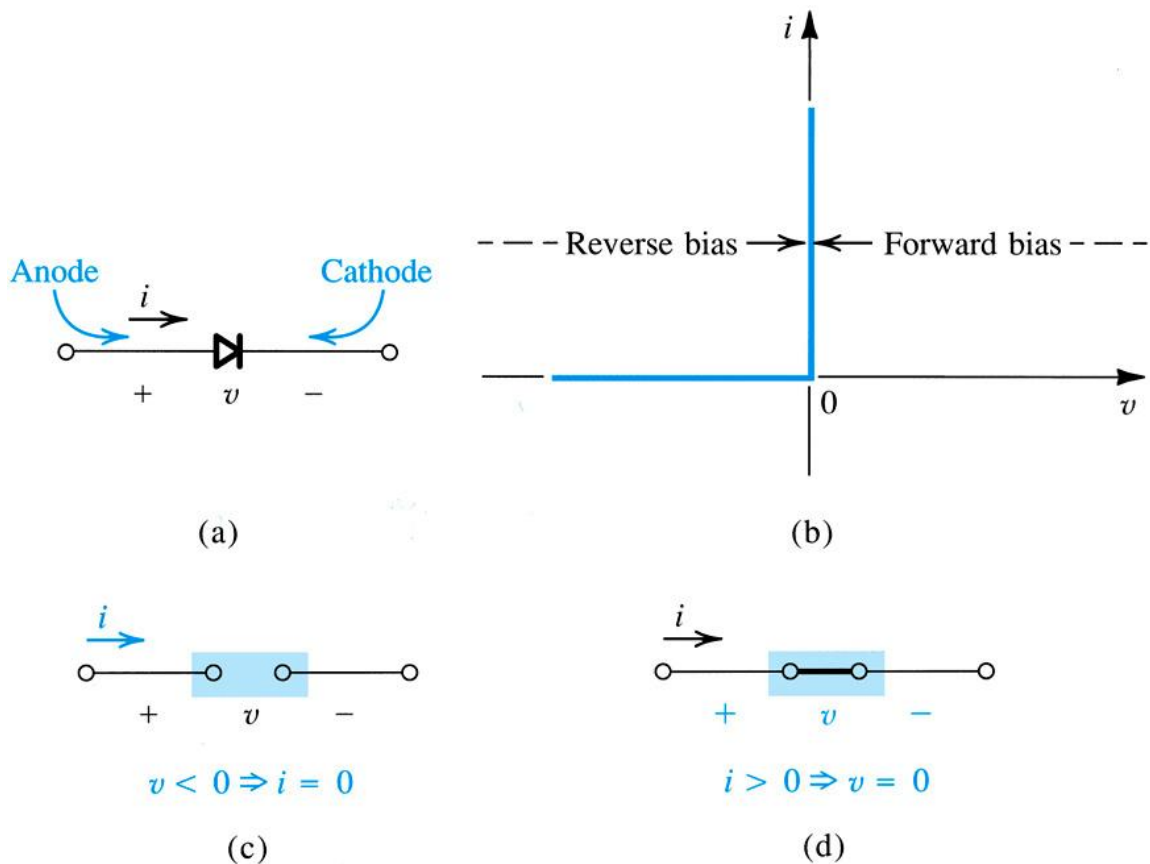
$$\frac{V_o(s)}{V_i(s)} = \frac{-R_2/R_1}{1 + sC_2R_2}$$

$$\omega_0 = \frac{1}{C_2R_2}$$



ideal diode

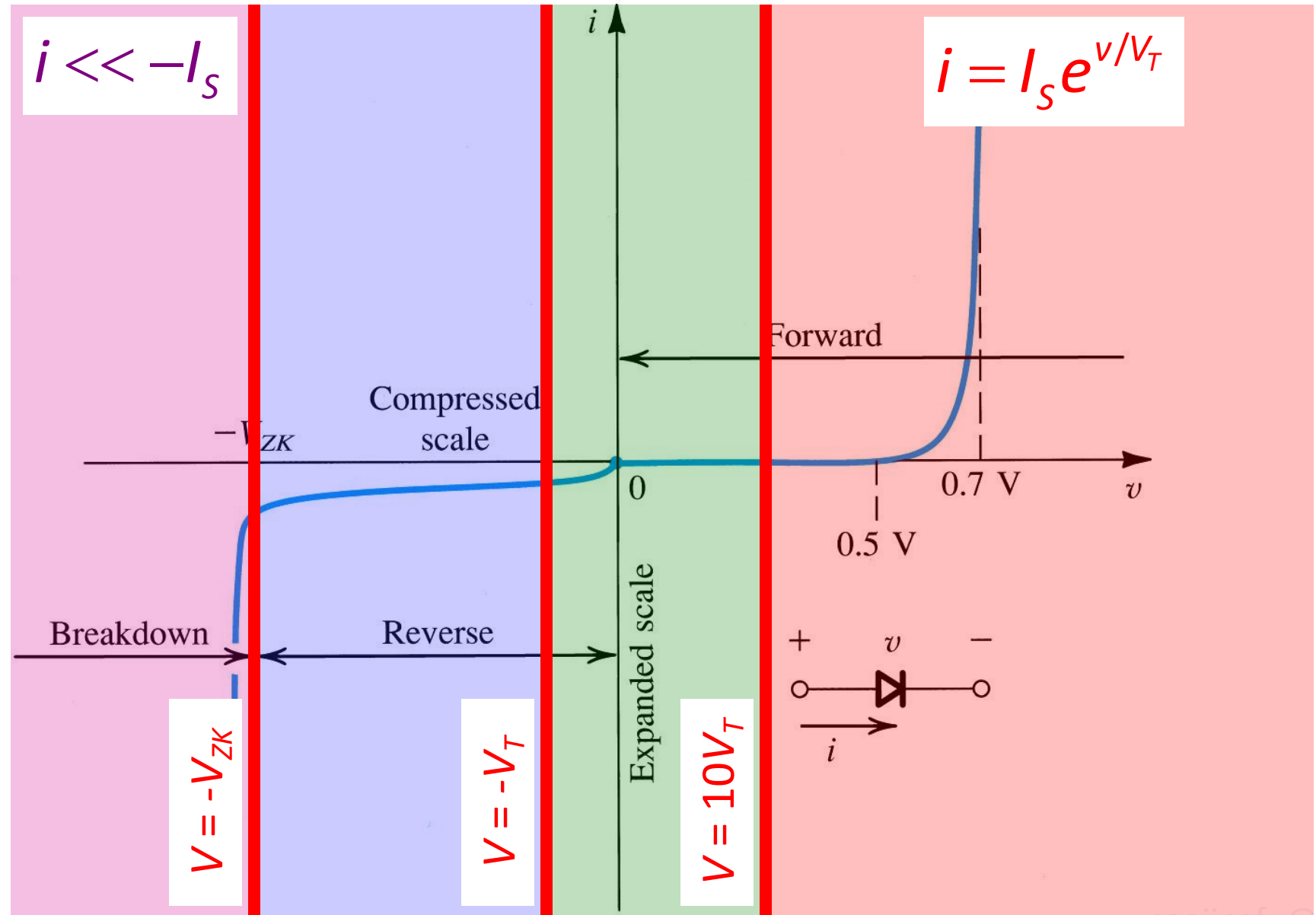
Chapter #4: Diodes



One **fundamental application** of this piecewise linear behavior is the **rectifier**

$$i = I_s (e^{v/V_T} - 1)$$

practical diode



The Exponential Model

example 4.4

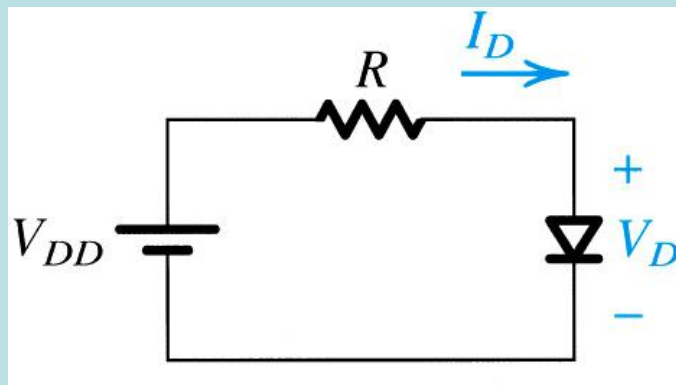
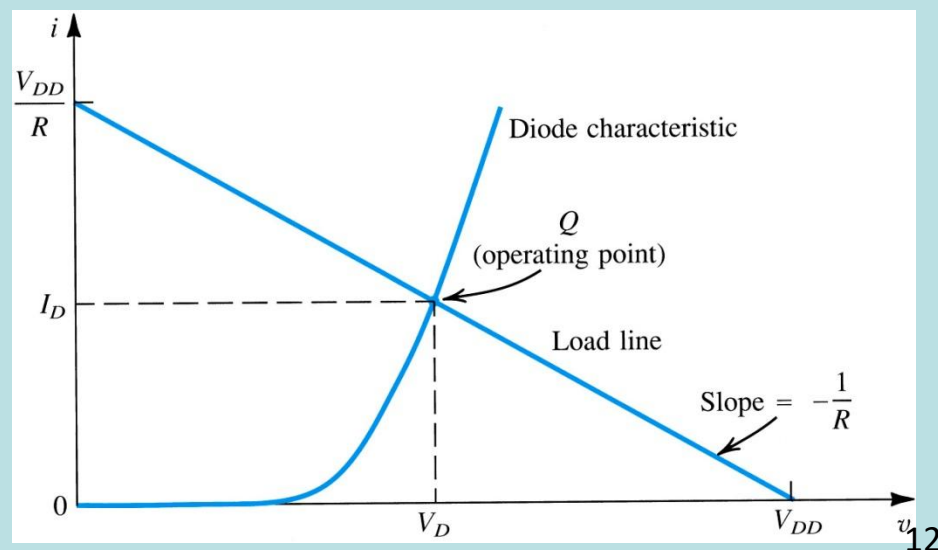


Figure 4.10: A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

$$(eq4.7) \quad I_D = \frac{V_{DD} - V_D}{R}$$



The Constant Voltage-Drop Model

- The **constant voltage-drop diode model** assumes that the slope of I_D vs. V_D is vertical @ 0.7V

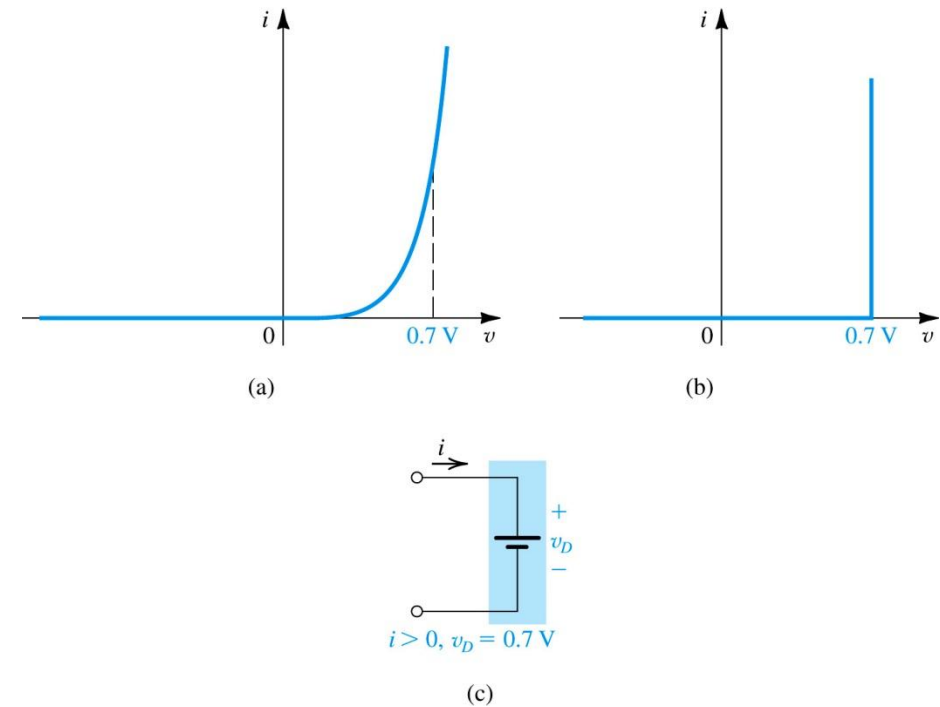


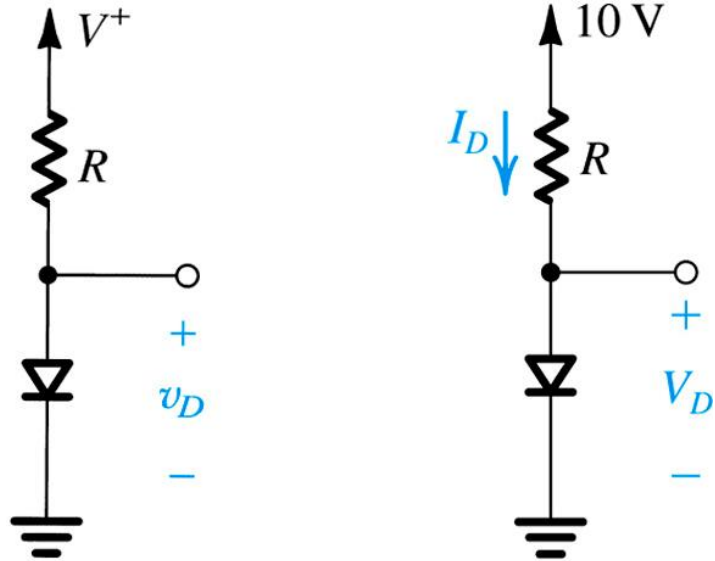
Figure 4.12: Development of the diode constant-voltage-drop model: (a) the...



When to use these models?

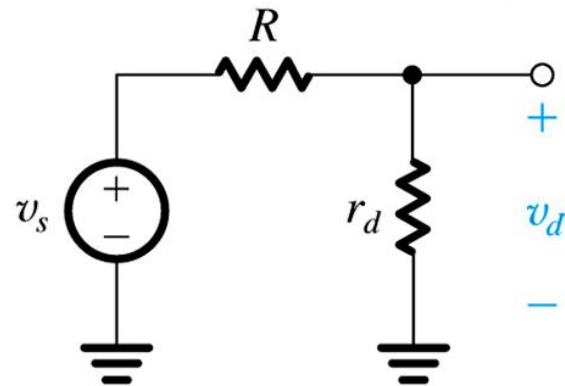
- **exponential model**
 - low voltages
 - less complex circuits
 - emphasis on accuracy over practicality
- **constant voltage-drop mode:**
 - medium voltages = $0.7V$
 - more complex circuits
 - emphasis on practicality over accuracy
- **ideal diode model**
 - high voltages $\gg 0.7V$
 - very complex circuits
 - cases where a difference in voltage by $0.7V$ is negligible
- **small-signal model**
 - this is next...

Small-Signal Model



(a)

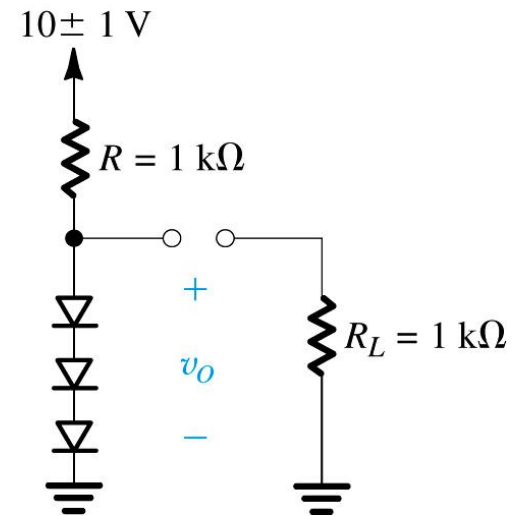
(b)

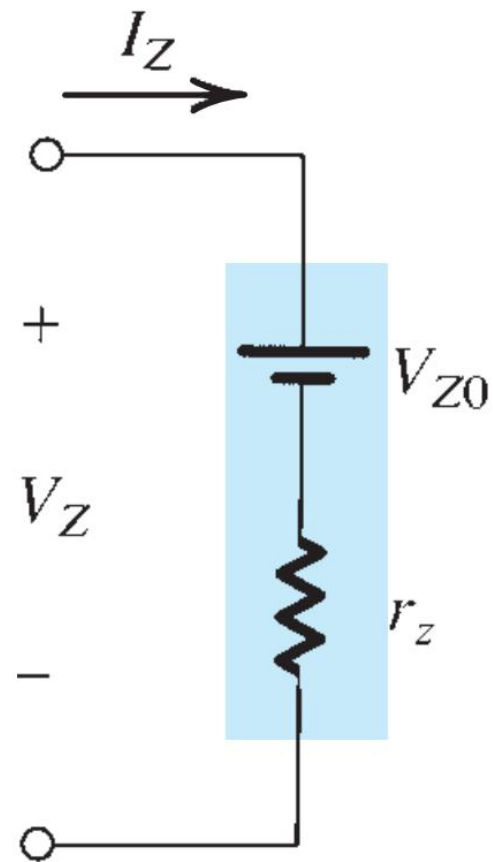
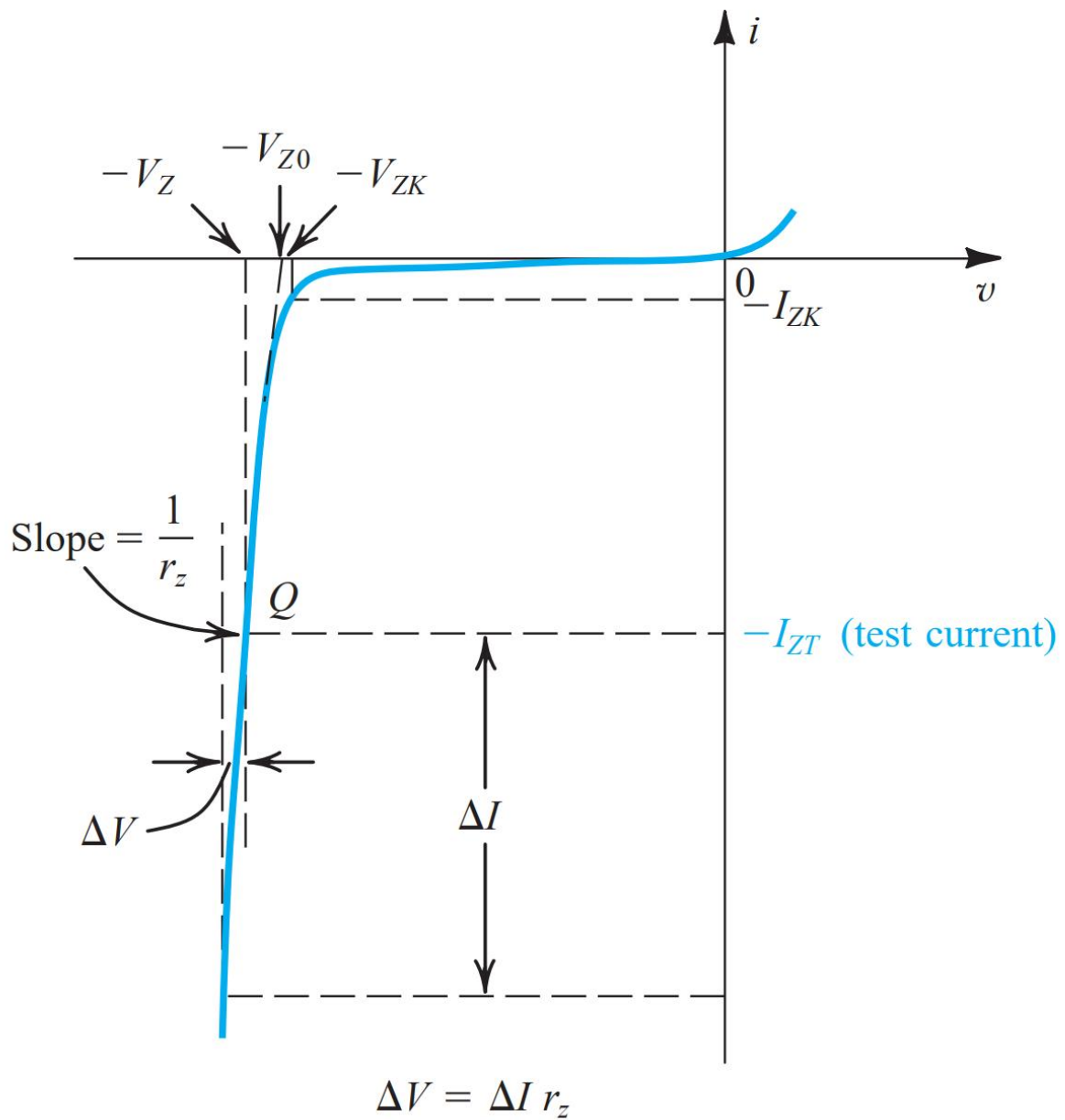


(c)

$$r_d = \frac{V_T}{I_D}$$

Application: Diode-Based Voltage Regulator





$$V_Z = V_{Z0} + r_z I_Z$$

The 6.8-V zener diode in the circuit of Fig. 4.19(a) is specified to have $V_Z = 6.8$ V at $I_Z = 5$ mA, $r_z = 20$ Ω , and $I_{ZK} = 0.2$ mA. The supply voltage V^+ is nominally 10 V but can vary by ± 1 V.

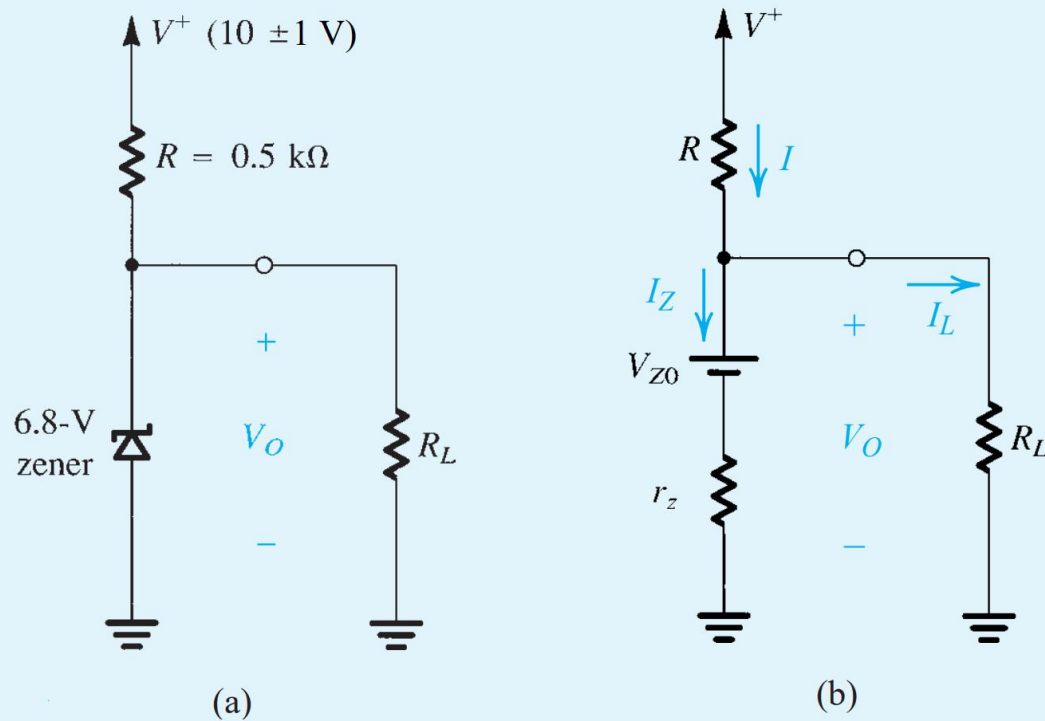


Figure 4.19 (a) Circuit for Example 4.7. (b) The circuit with the zener diode replaced with its equivalent circuit model.

- Find V_O with no load and with V^+ at its nominal value.
- Find the change in V_O resulting from the ± 1 -V change in V^+ . Note that $(\Delta V_O / \Delta V^+)$, usually expressed in mV/V, is known as **line regulation**.
- Find the change in V_O resulting from connecting a load resistance R_L that draws a current $I_L = 1$ mA, and hence find the **load regulation** $(\Delta V_O / \Delta I_L)$ in mV/mA.
- Find the change in V_O when $R_L = 2$ k Ω .
- Find the value of V_O when $R_L = 0.5$ k Ω .
- What is the minimum value of R_L for which the diode still operates in the breakdown region?

$$I_Z = I = \frac{V^+ - V_{Z0}}{R + r_z}$$

$$= \frac{10 - 6.7}{0.5 + 0.02} = 6.35 \text{ mA}$$

$$V_O = V_{Z0} + I_Z r_z$$

$$= 6.7 + 6.35 \times 0.02 = 6.83 \text{ V}$$

$$\Delta V_O = \Delta V^+ \frac{r_z}{R + r_z}$$

$$= \pm 1 \times \frac{20}{500 + 20} = \pm 38.5 \text{ mV}$$

$$\text{Line regulation} = 38.5 \text{ mV/V}$$

$$\Delta V_O = r_z \Delta I_Z$$

$$= 20 \times -1 = -20 \text{ mV}$$

$$\text{Load regulation} \equiv \frac{\Delta V_O}{\Delta I_L} = -20 \text{ mV/mA}$$

$$V_O = V^+ \frac{R_L}{R + R_L}$$

$$= 10 \frac{0.5}{0.5 + 0.5} = 5 \text{ V}$$

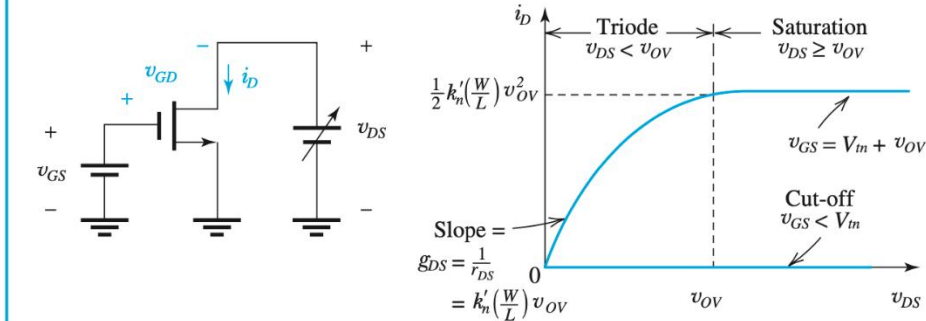
$$R_L = \frac{6.7}{4.4} \approx 1.5 \text{ k}\Omega$$

Chapter #5: MOSFET's

Table 5.1. provides a
**compilation of the
 conditions and
 formulas** for operation
 of NMOS transistor in
 three regions.

**cutoff
 triode
 saturation**

Table 5.1 Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{in}$: no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{in} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{in}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_{in}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

or equivalently,

$$i_D = k'_n \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2} v_{DS} \right) v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{in}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

Then

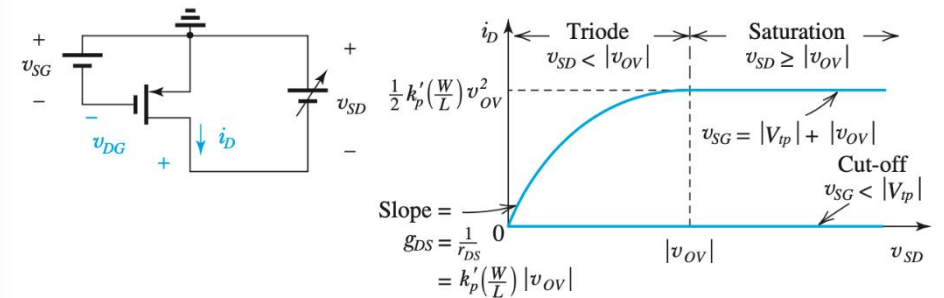
$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{in})^2$$

or equivalently,

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

MOSFET Circuits at DC

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cut-off; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently:

$$v_{SD} < |v_{OV}|$$

Then,

$$i_D = k'_p \left(\frac{W}{L} \right) \left[(v_{SG} - |V_{tp}|) v_{SD} - \frac{1}{2} v_{SD}^2 \right]$$

or equivalently

$$i_D = k'_p \left(\frac{W}{L} \right) \left(|v_{OV}| - \frac{1}{2} v_{SD} \right) v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2} k'_p \left(\frac{W}{L} \right) (v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2} k'_p \left(\frac{W}{L} \right) v_{OV}^2$$

- MOSFET's DC (bias, Q) points can be obtained by Table 5.1&5.2.
- We will neglect the effects of channel length modulation (assuming $\lambda = 0$).
- We will work in terms of overdrive voltage (v_{OV}), which reduces need to distinguish between PMOS and NMOS.
- Sometime quadrature equations should be solved.

Small signal model

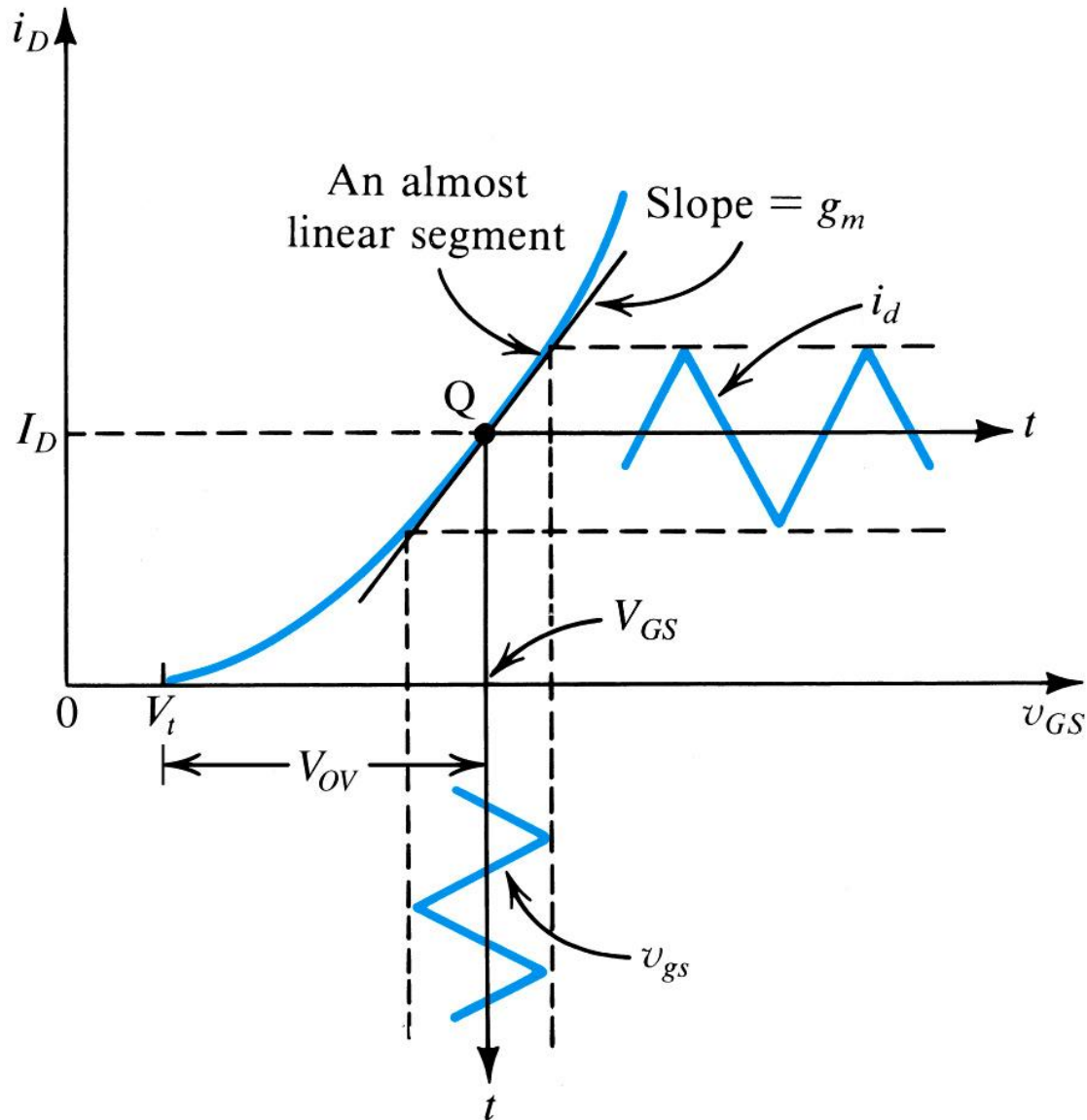


Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET

Small-Signal Parameters

NMOS transistors

Transconductance:

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{OV} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{V_{OV}}$$

$$(eq5.55) \quad g_m = k'_n \frac{W}{L} V_{OV}$$

$$(eq5.56) \quad g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

Output resistance:

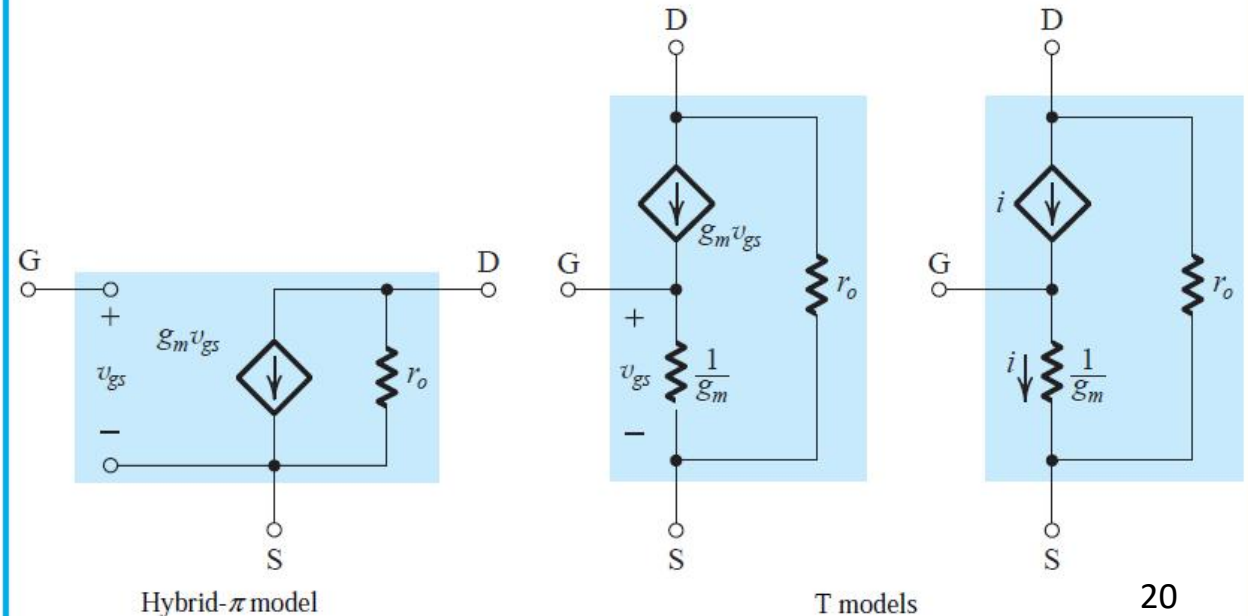
$$r_o = V_A / I_D = 1 / \lambda I_D$$

$$(eq5.57) \quad g_m = \frac{2I_D}{V_{OV}}$$

PMOS transistors

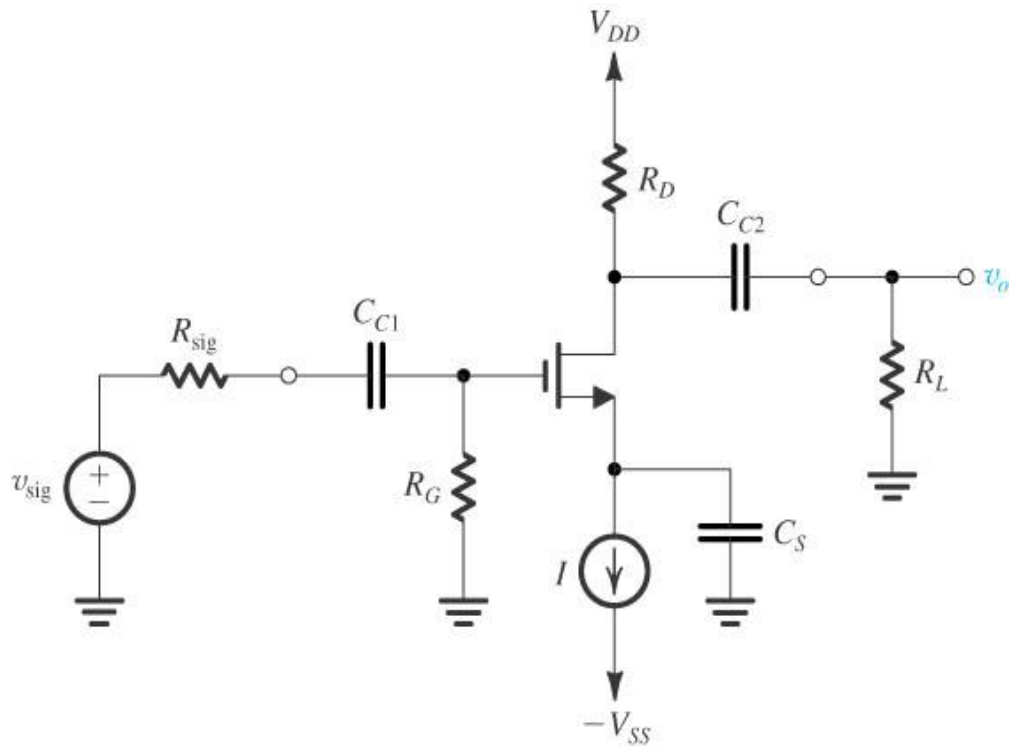
Same formulas as for NMOS except using $|V_{OV}|$, $|V_A|$, and replacing μ_n with μ_p .

Small-Signal Equivalent Circuit Models

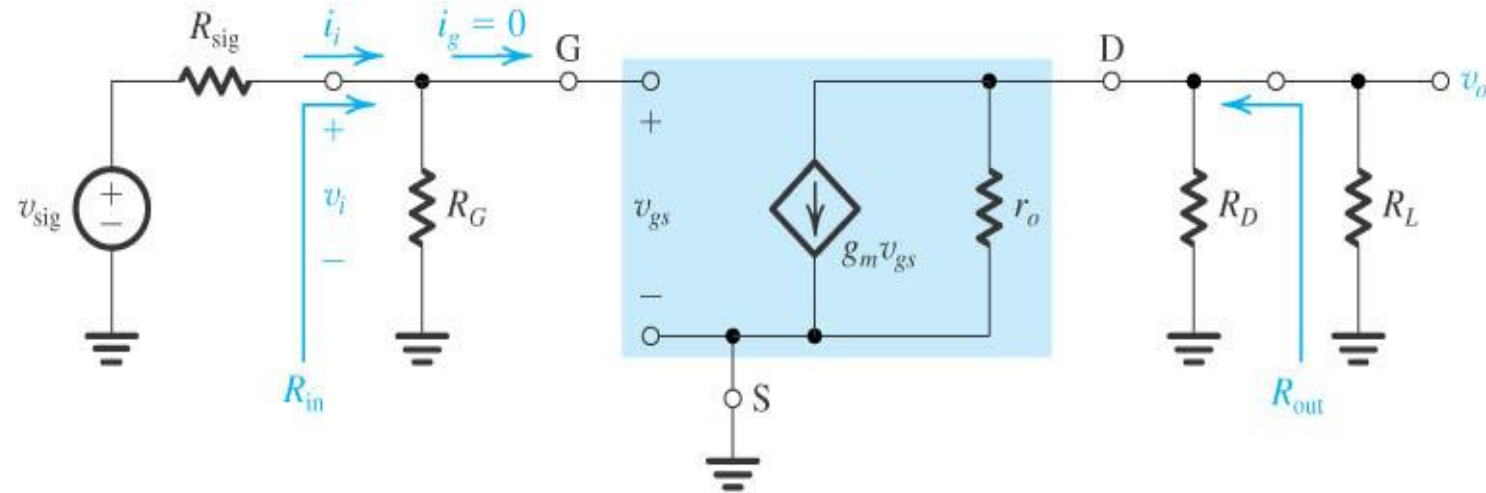


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Figure 5.35: Small-signal operation of the MOSFET amplifier.

The Common-Source Amplifier



(a)



(b)

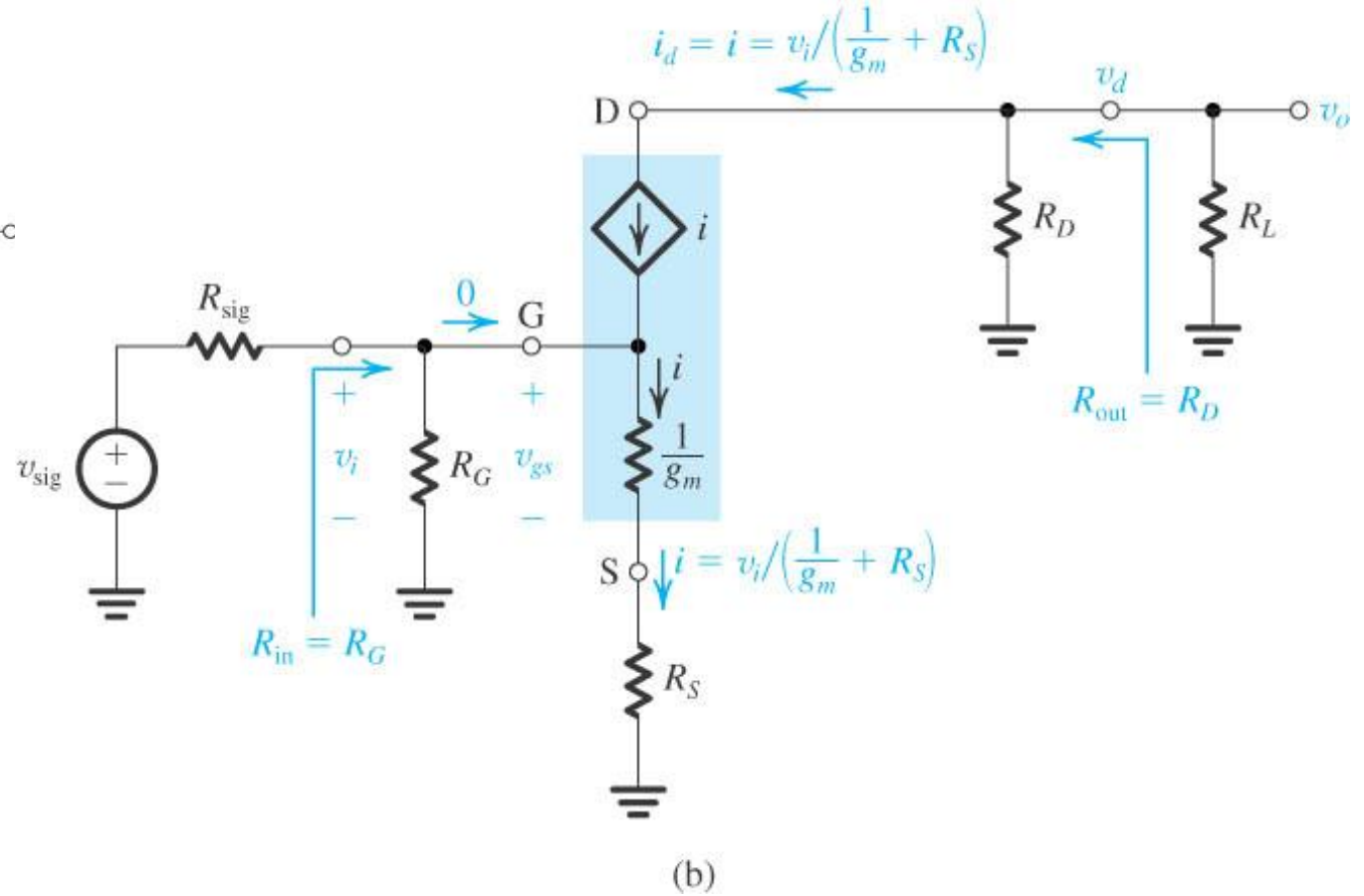
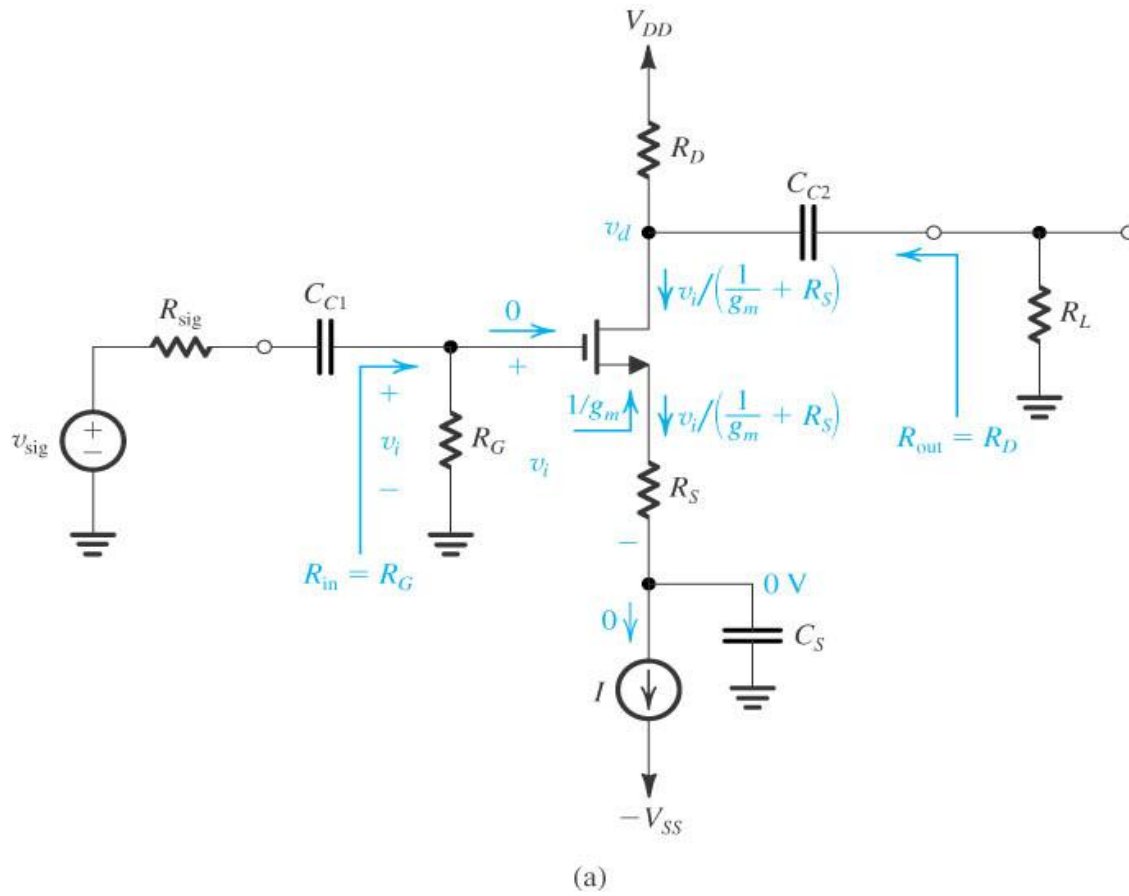
$$R_{in} = R_G \quad R_{out} = r_o \parallel R_D$$

$$A_v = -g_m (r_o \parallel R_D \parallel R_L)$$

$$G_v = -\frac{R_G}{R_G + R_{sig}} g_m (R_D \parallel R_L \parallel r_o)$$

- Very high input resistance
- Moderately high voltage gain
- Relatively high output resistance

The Common-Source Amplifier with a Source Resistance

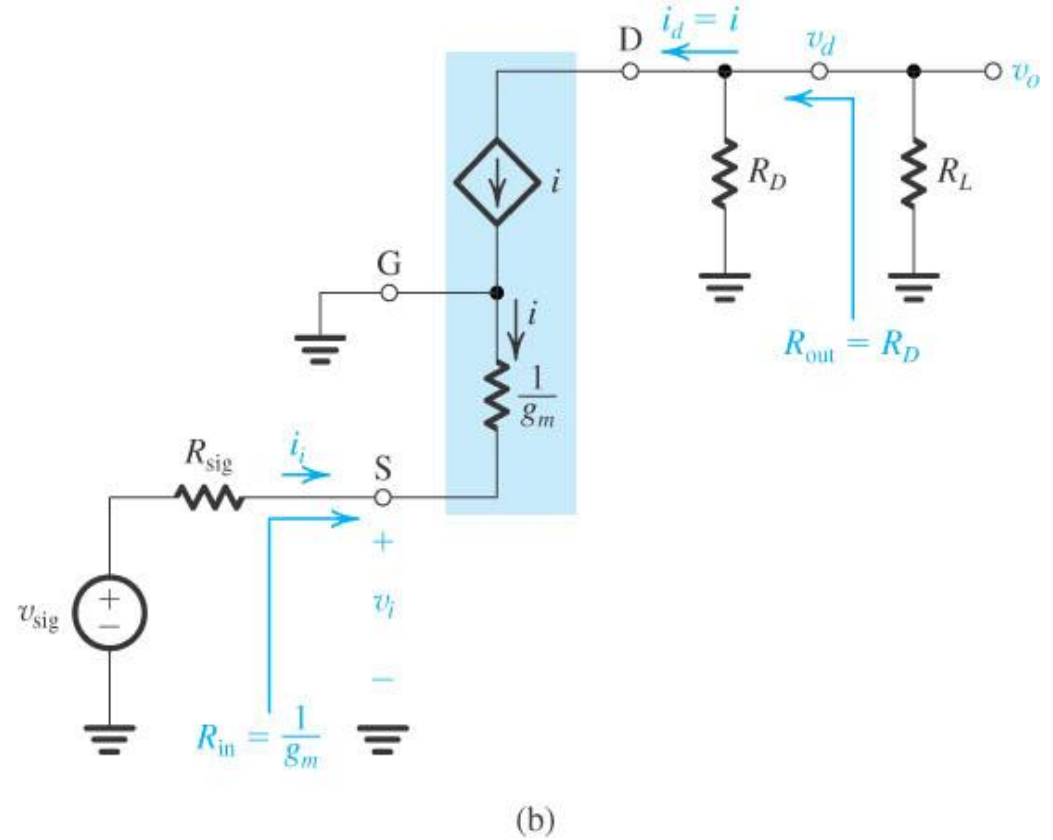
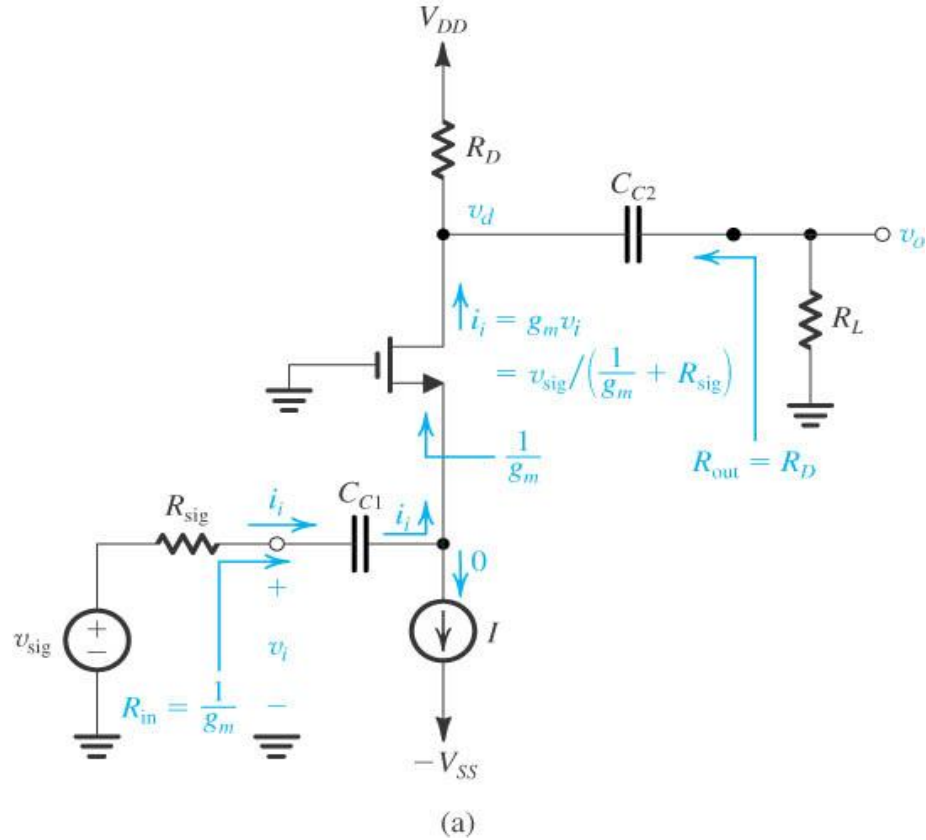


$$R_{in} = R_G \quad A_v = -\frac{g_m (R_D // R_L)}{1 + g_m R_S}$$

$$R_{out} = R_D \quad G_v = -\frac{R_G}{R_G + R_{sig}} \frac{g_m (R_D // R_L)}{1 + g_m R_S}$$

- Including R_S results in a gain reduction by the factor $(1 + g_m R_S)$
- R_S takes the effect of negative feedback.

The Common-Gate Amplifier



$$R_{in} = \frac{1}{g_m}$$

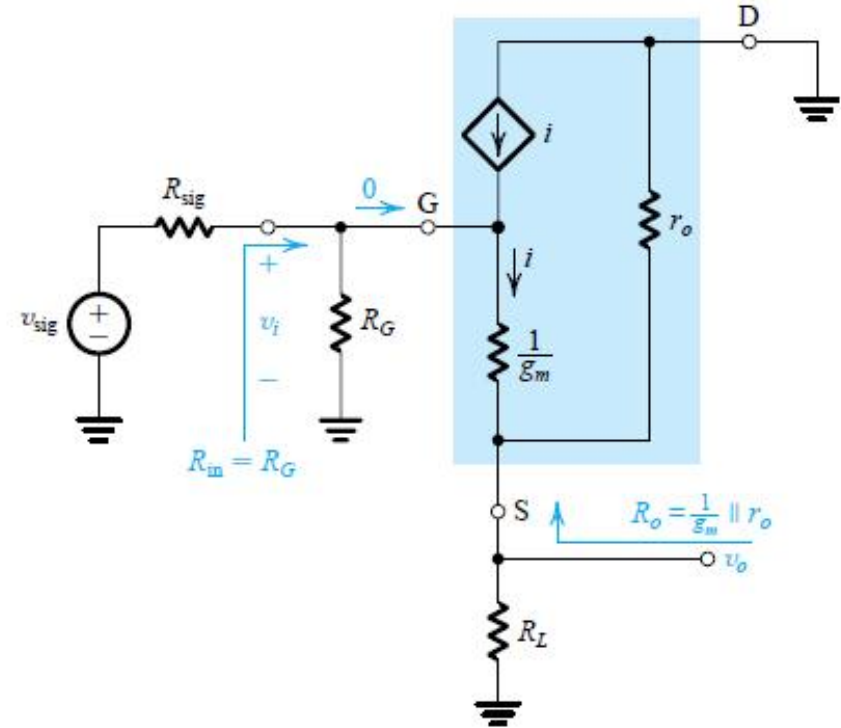
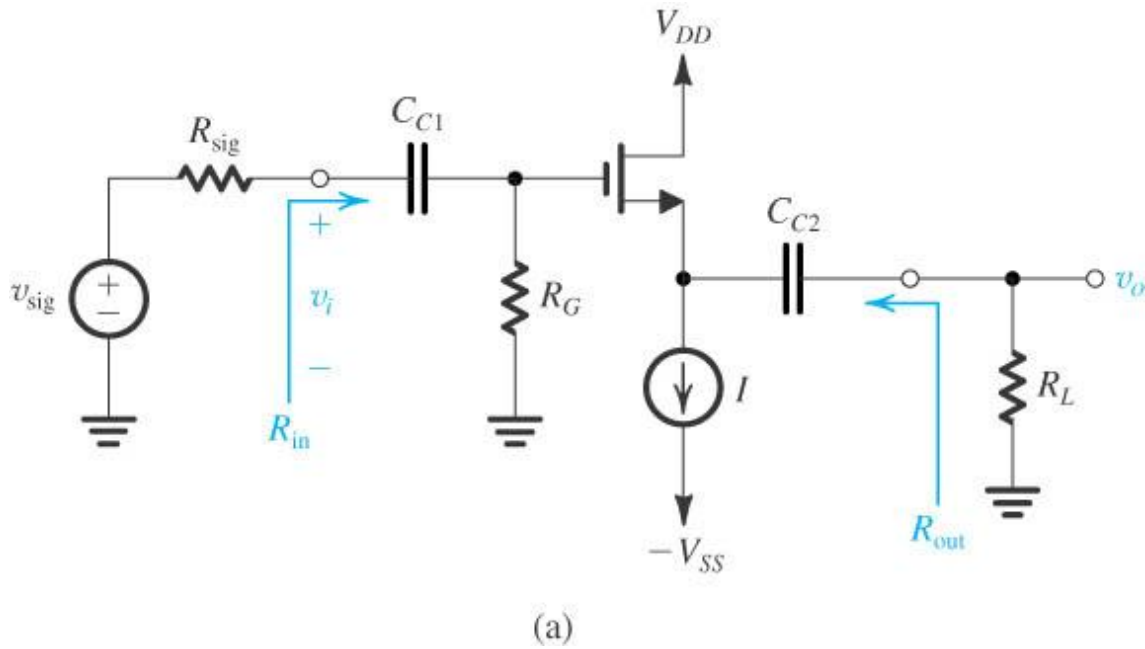
$$R_{out} = R_D$$

$$A_v = g_m (R_D // R_L) \quad G_v = \frac{g_m (R_D // R_L)}{1 + g_m R_{sig}}$$

2026-5-23

- Noninverting amplifier
- Low input resistance
- Nearly identical voltage gain of CS amplifier
- overall voltage gain is smaller by the factor $(1 + g_m R_{sig})$
- Relatively high output resistance
- Current follower

The Common-Drain or Source-Follower Amplifier

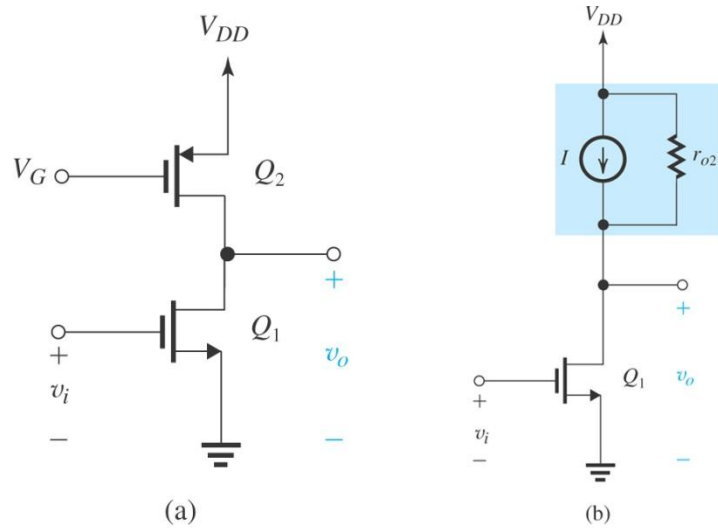


$$R_{in} = R_G \quad R_{out} = \frac{1}{g_m} \parallel r_o \cong \frac{1}{g_m}$$

$$A_v = \frac{r_o \parallel R_L}{r_o \parallel R_L + \frac{1}{g_m}} \quad G_v = \frac{R_G}{R_G + R_{sig}} \frac{r_o \parallel R_L}{r_o \parallel R_L + \frac{1}{g_m}} \approx 1$$

- Very high input resistance
- Voltage gain is close to unity
- Relatively low output resistance
- Voltage buffer amplifier
- Power amplifier

Chapter #7: Building Blocks of Integrated Circuit Amplifiers



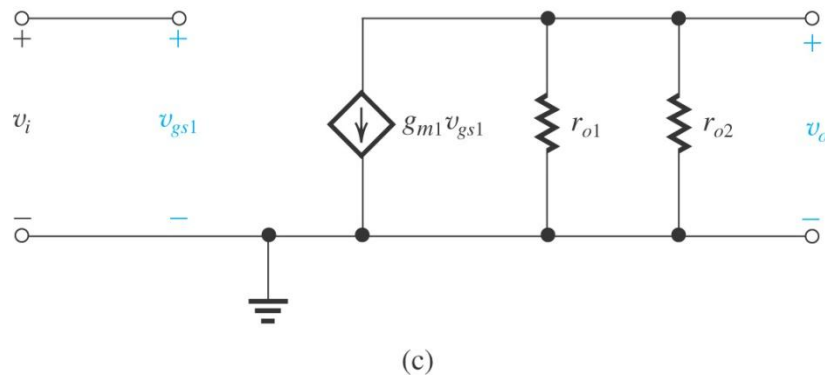
for PMOS implementation of active-loaded CS amplifier:

large-signal MOSFET model

$$(7.16) \quad I = \frac{1}{2} (\mu_p C_{ox}) \frac{W}{L} [V_{DD} - V_G - |V_{tp}|]^2$$

$$(7.17) \quad r_{o2} = \frac{|V_{A2}|}{I}$$

current-source
no longer has
infinite output
resistance



$$(7.18) \quad A_v = \frac{v_o}{v_i} = -g_{m1} (r_{o1} || r_{o2})$$

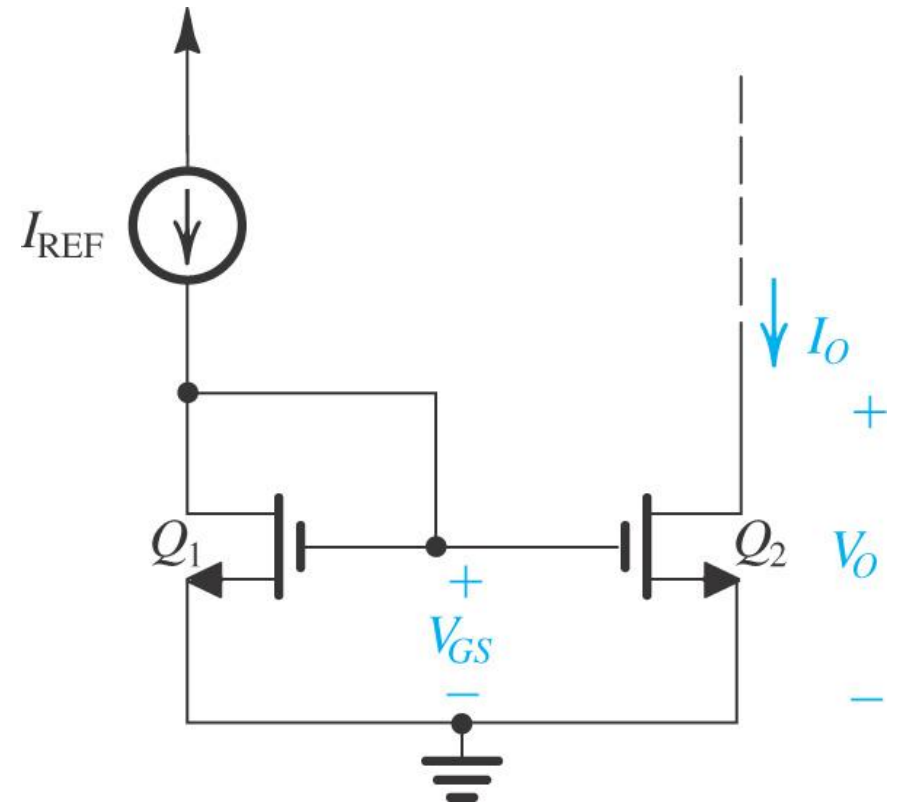
The Basic MOSFET Current Source

$$(7.52) \quad I_{D1} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_1 (V_{GS} - V_{tn})^2$$

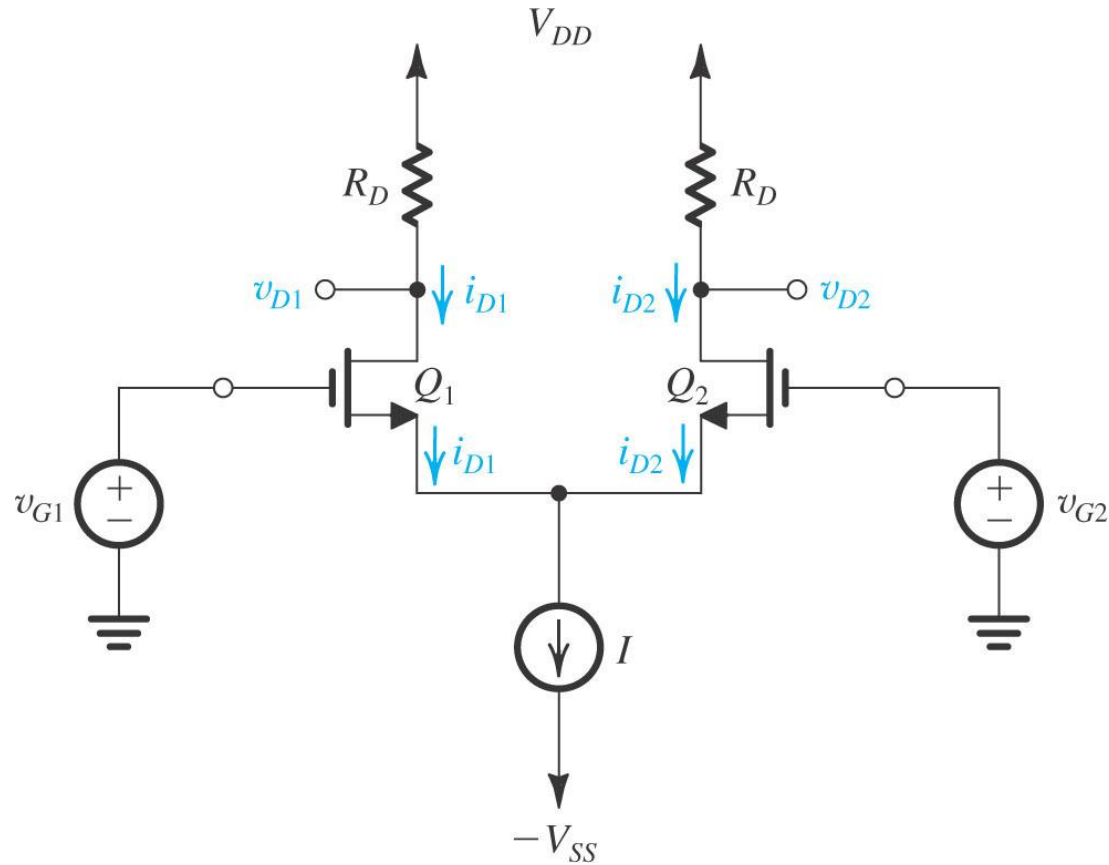
$$(7.53) \quad I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R}$$

$$(7.54) \quad I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_2 (V_{GS} - V_{tn})^2$$

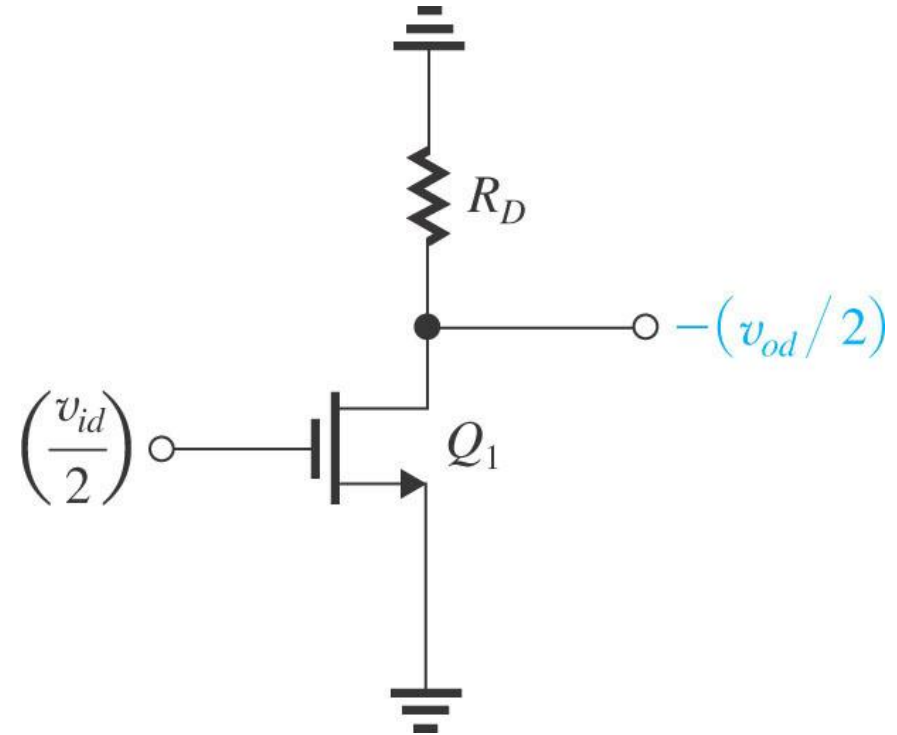
$$(7.55) \quad \frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$



Chapter #8: Differential and Multistage Amplifiers

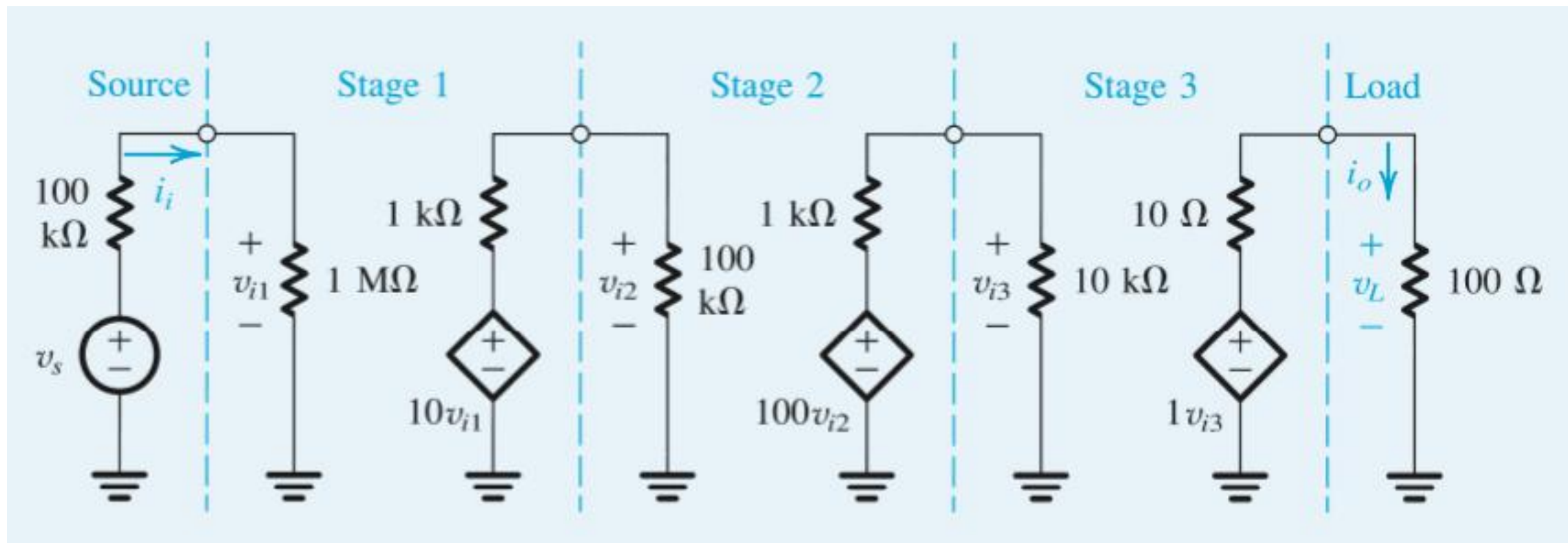


The MOS Differential Pair

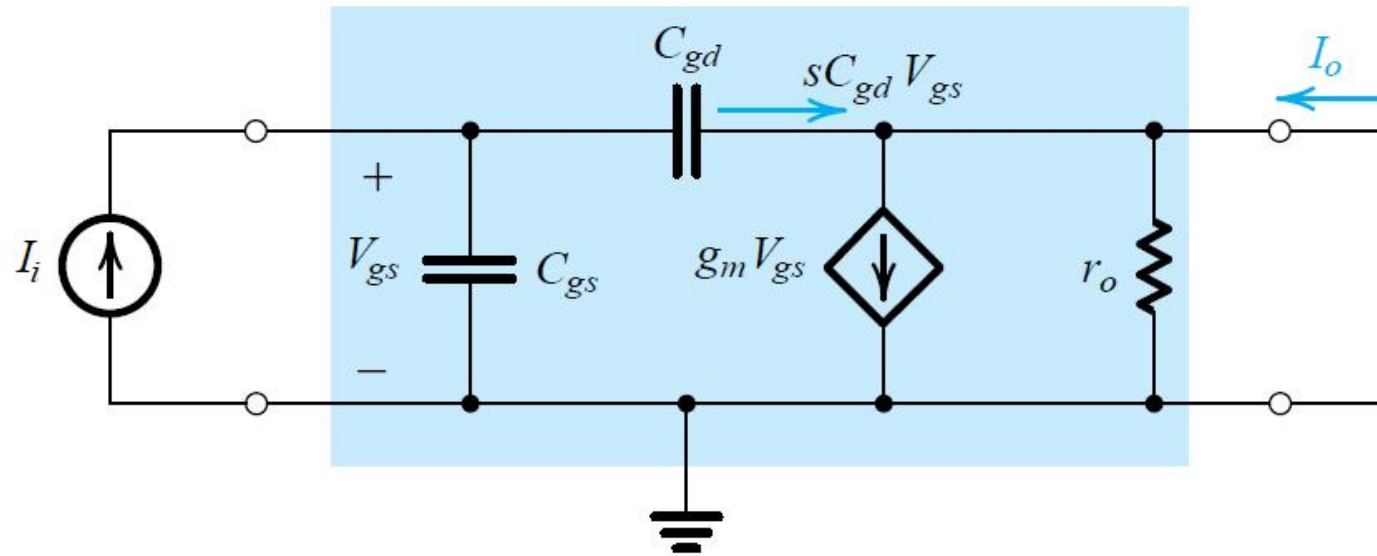


The Differential Half-Circuit

Multi-stage amplifier



Chapter #9: Frequency Response



internal capacitances – cause the falloff of gain at high frequencies like those seen in previous example

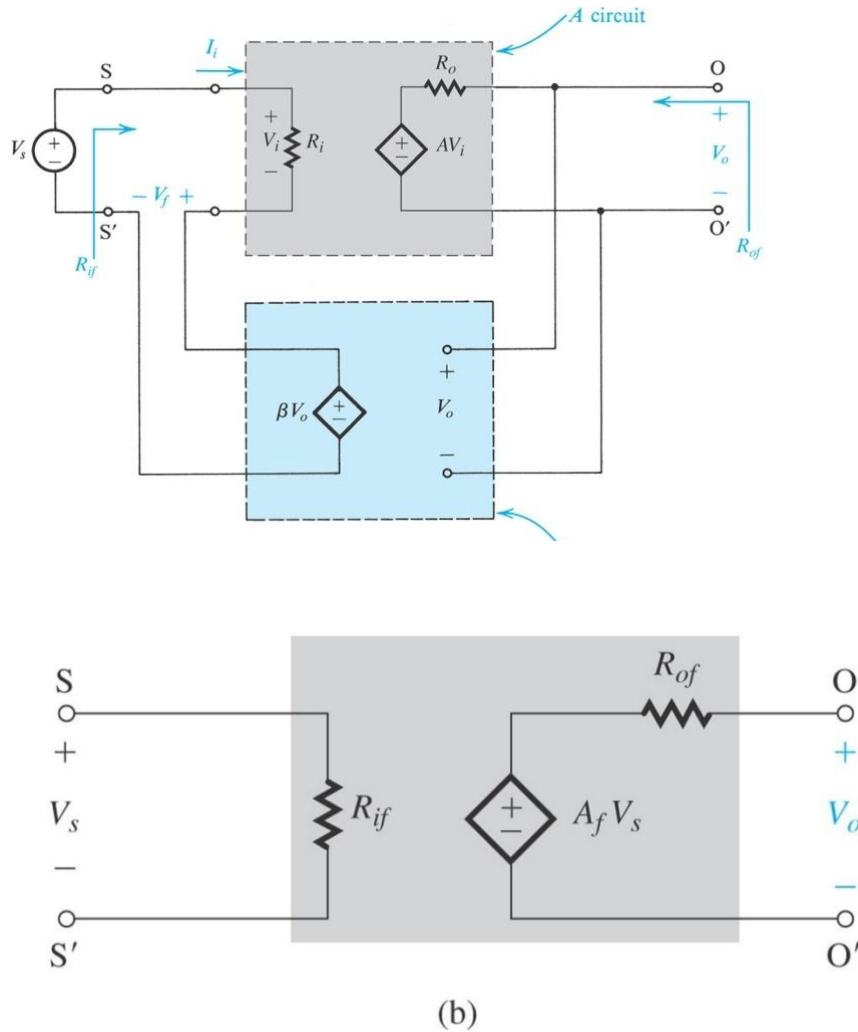
coupling capacitors – cause the falloff of gain at low frequencies are placed in between amplifier stages generally chosen to be large

$$(9.28) \quad I_o \approx g_m V_{gs} \quad (9.29) \quad V_{gs} = I_i / j\omega(C_{gs} + C_{gd})$$

$$(9.30) \text{ short circuited current-gain: } \frac{I_o}{I_i} = \frac{g_m}{j\omega(C_{gs} + C_{gd})}$$

$$(9.31) \text{ unity-gain frequency: } f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$

Ideal feedback voltage amplifier



(10.17) closed-loop gain: $A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$

(10.18) input current: $I_i = \frac{V_i}{R_i} = \frac{V_s}{(1 + A\beta)R_i}$

(10.19) input resistance: $R_{if} = (1 + A\beta)R_i$

(10.20) output resistance: $R_{of} = \frac{V_x}{I_x}$

(10.21) current-x: $I_x = \frac{V_x - AV_i}{R_o}$

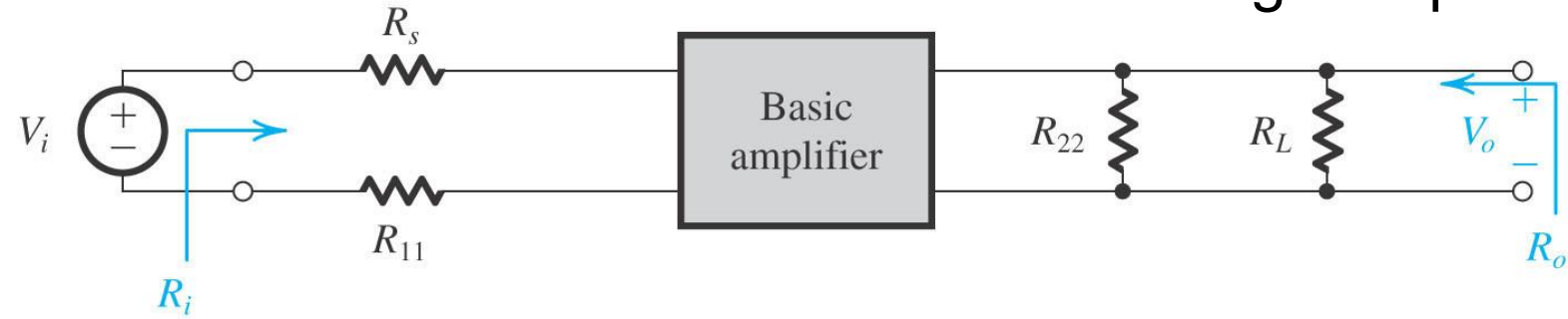
(10.22) output resistance: $R_{of} = \frac{R_o}{1 + A\beta}$

Figure 10.12: The **series-shunt** feedback amplifier: (a) ideal structure; (b) equivalent circuit.

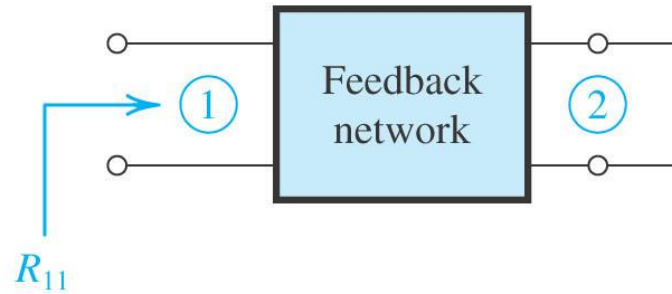
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Practical feedback voltage amplifier

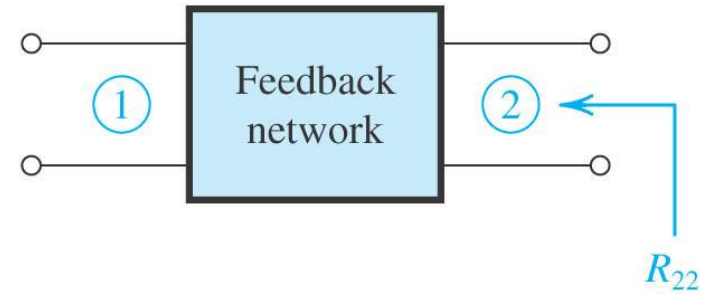
(a) The A circuit is



where R_{11} is obtained from

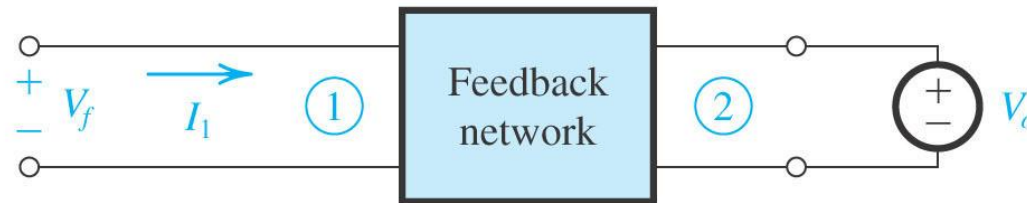


and R_{22} is obtained from



and the gain A is defined $A \equiv \frac{V_o}{V_i}$

(b) β is obtained from



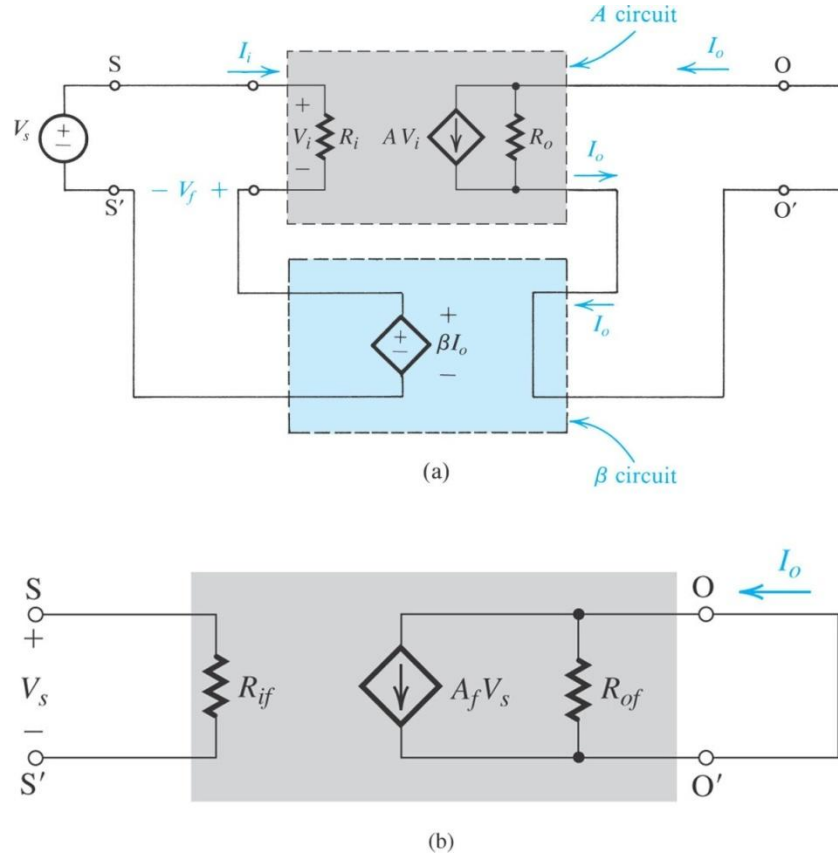
$$\beta \equiv \left. \frac{V_f}{V_o} \right|_{I_1 = 0}$$

$$R_{in} = R_{if} - R_s \quad (10.24)$$

$$R_{out} = 1 / \left(\frac{1}{R_{of}} - \frac{1}{R_L} \right) \quad (10.25)$$

Figure 10.15 Summary of the rules for finding the A circuit and β for the series-shunt case of Fig. 10.14(a).

Ideal Feedback Transconductance Amplifier



$$(10.26) \quad A_f = \frac{I_o}{V_s} = \frac{A}{1 + A\beta}$$

$$(10.27) \quad R_{if} = (1 + A\beta)R_i$$

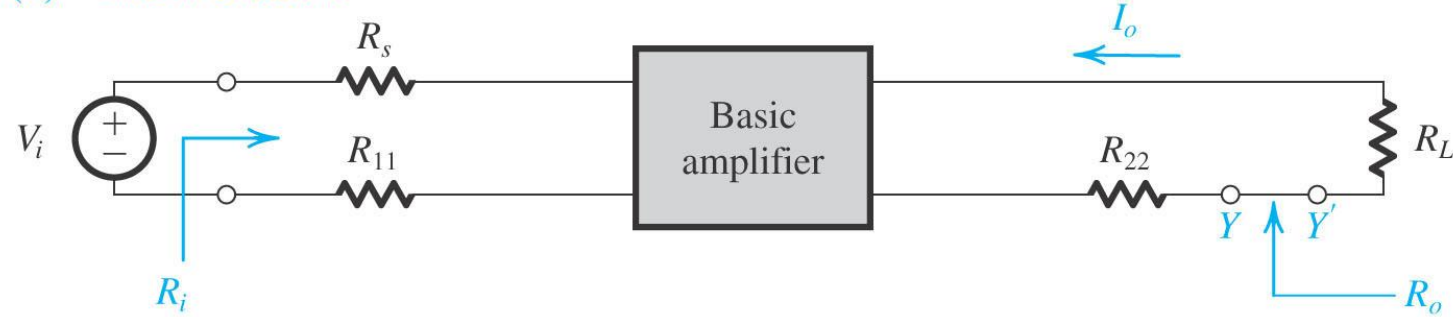
$$(10.29) \quad R_{of} = (1 + A\beta)R_o$$

Figure 10.18: The **series-series** feedback amplifier: (a) ideal structure; (b) equivalent circuit.

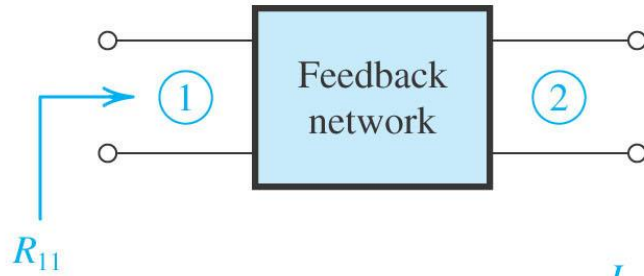
Figure 10.21 Finding the A circuit and β for the series-series feedback amplifier.

Practical Feedback Transconductance Amplifier

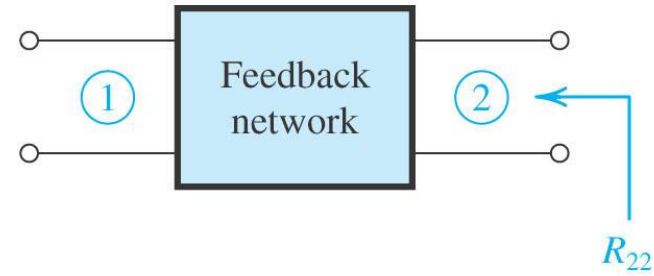
(a) The A circuit is



where R_{11} is obtained from

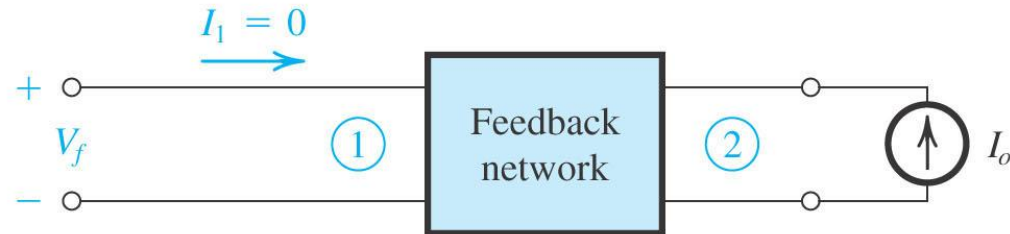


and R_{22} is obtained from



and the gain A is defined $A \equiv \frac{I_o}{V_i}$

(b) β is obtained from

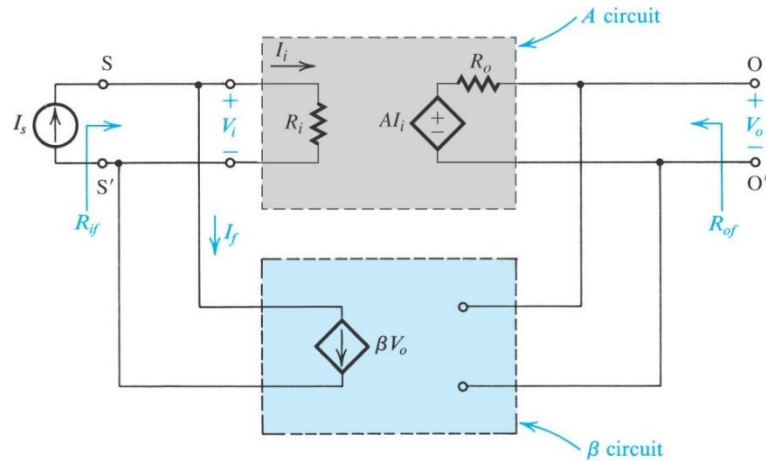


$$\beta \equiv \left. \frac{V_f}{I_o} \right|_{I_i = 0}$$

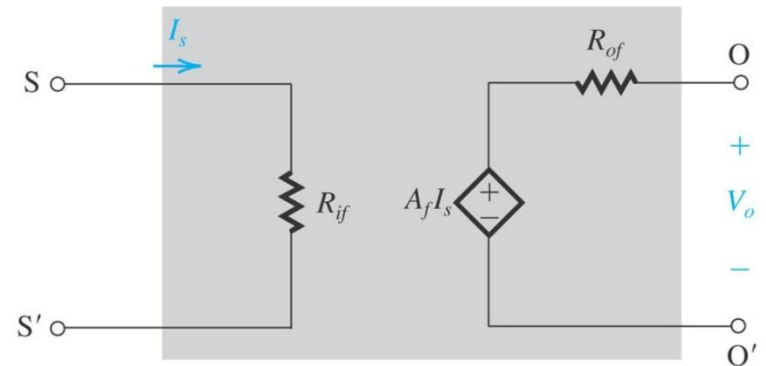
$$R_{in} = R_{if} - R_s \quad (10.31)$$

$$R_{out} = R_{of} - R_L \quad (10.32)$$

Ideal Feedback Transresistance Amplifier



(a)



(b)

$$(10.43) A_f = \frac{V_o}{I_s} = \frac{A}{1 + A\beta}$$

$$(10.44) R_{if} = \frac{R_i}{1 + A\beta}$$

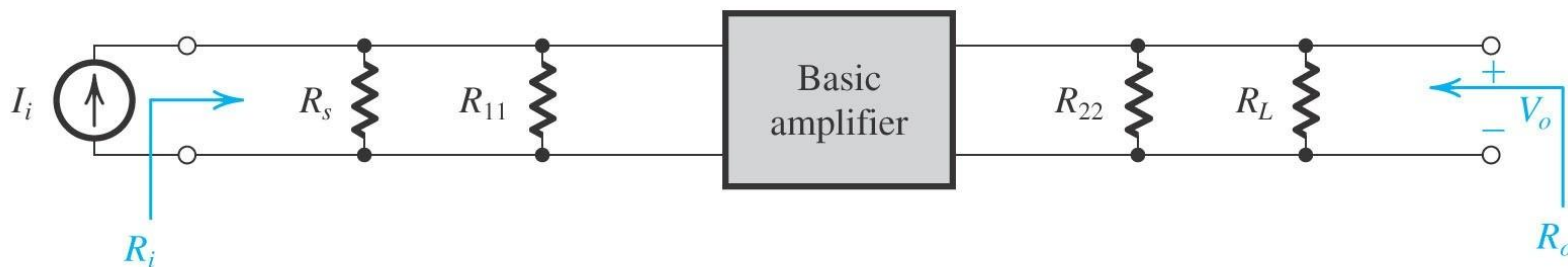
$$(10.45) R_{of} = \frac{R_o}{1 + A\beta}$$

Figure 10.24: (a) Ideal structure for the **shunt-shunt** feedback amplifier. (b) Equivalent circuit of the amplifier in (a).

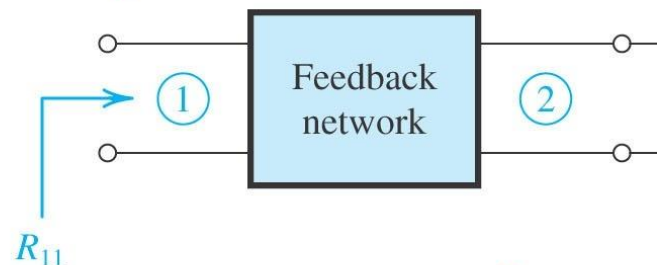
Practical Feedback Transresistance Amplifier

Figure 10.26 Finding the A circuit and β for the shunt-shunt feedback amplifier in Fig. 10.25.

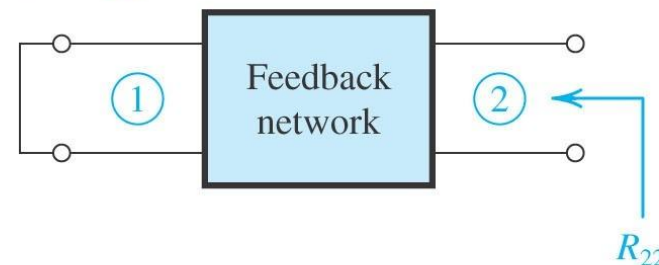
(a) The A circuit is



where R_{11} is obtained from

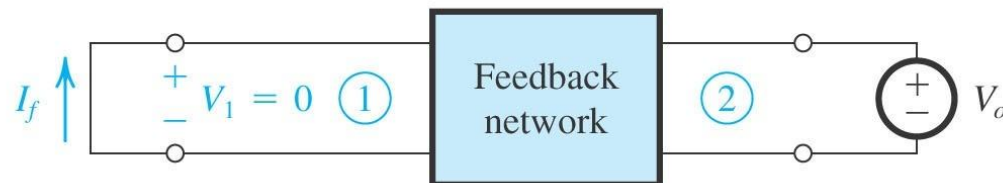


and R_{22} is obtained from



and the gain A is defined $A \equiv \frac{V_o}{I_i}$

(b) β is obtained from

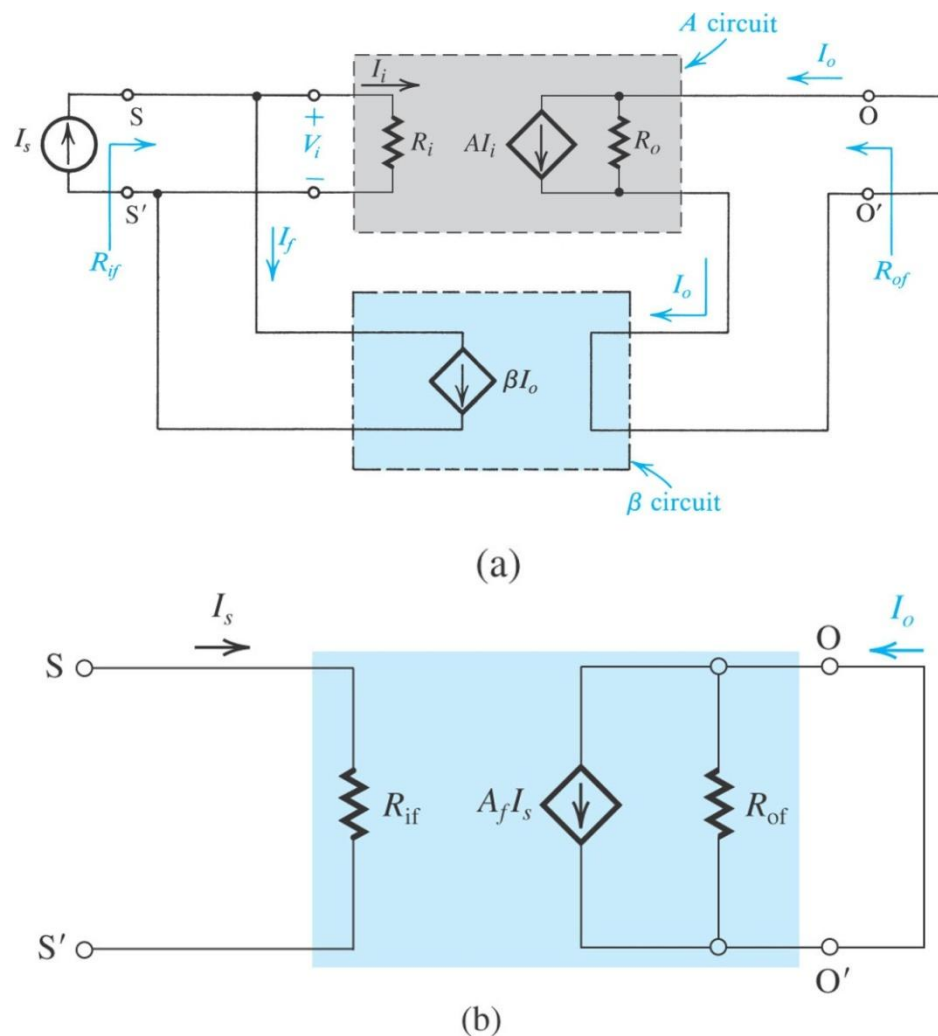


$$\beta \equiv \frac{I_f}{V_o} \Big|_{V_1 = 0}$$

$$R_{in} = 1 / \left(\frac{1}{R_{if}} - \frac{1}{R_s} \right) \quad (10.46)$$

$$R_{out} = 1 / \left(\frac{1}{R_{of}} - \frac{1}{R_L} \right) \quad (10.47)$$

Ideal Feedback Current Amplifier



$$(10.59) \quad A_f = \frac{I_o}{I_s} = \frac{A}{1 + A\beta}$$

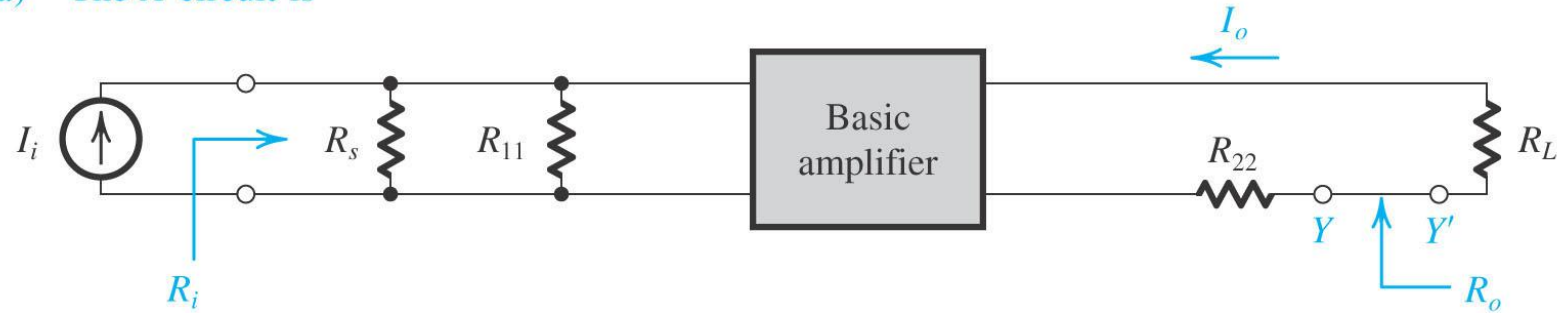
$$(10.60) \quad R_{if} = \frac{R_i}{1 + A\beta}$$

$$(10.61) \quad R_{of} = (1 + A\beta)R_o$$

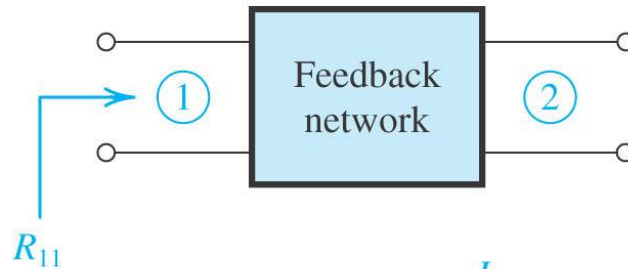
Figure 10.28: (a) Ideal structure for the **shunt-series** feedback amplifier. (b) Equivalent circuit of the amplifier in (a).

Practical Feedback Current Amplifier

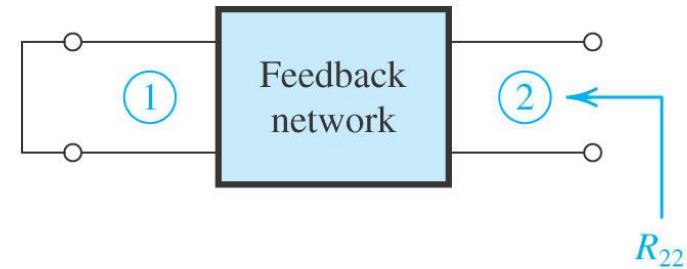
(a) The A circuit is



where R_{11} is obtained from

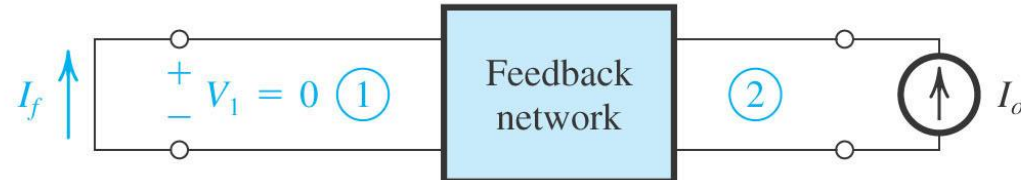


and R_{22} is obtained from



and the gain A is defined as $A \equiv \frac{I_o}{I_i}$

(b) β is obtained from



$$\beta \equiv \left. \frac{I_f}{I_o} \right|_{V_1 = 0}$$

$$R_{in} = 1 / \left(\frac{1}{R_{if}} - \frac{1}{R_s} \right) \quad (10.62)$$

$$R_{out} = R_{of} - R_L \quad (10.63)$$

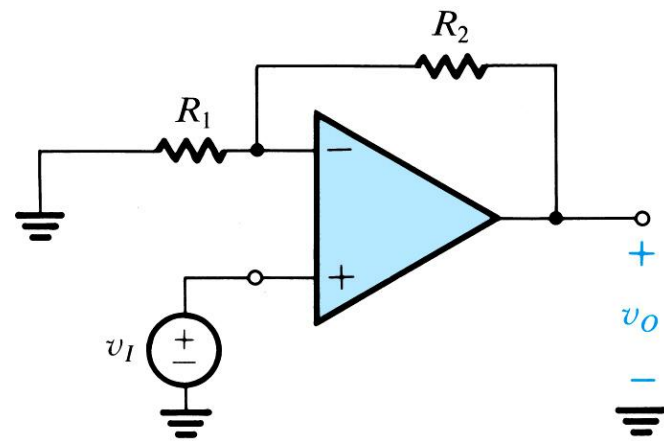
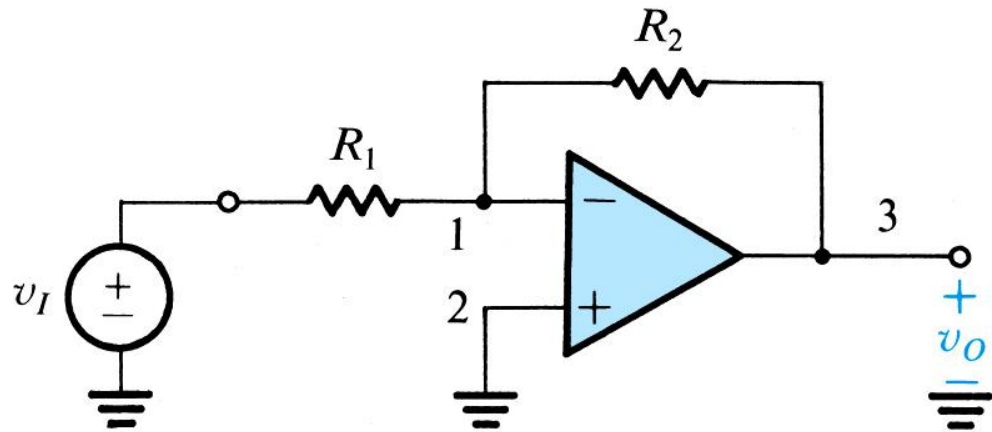
Figure 10.30 Finding the A circuit and β for the current-mixing current-sampling (shunt-series) feedback amplifier of Fig. 10.29.

Table 10.1 Summary of Relationships for the Four Feedback-Amplifier Topologies

Feedback Amplifier	Feedback Topology	x_i	x_o	x_f	x_s	A	β	A_f	Source Form	Loading of Feedback Network is Obtained		To Find β , Apply to Port 2 of Feedback Network	R_{if}	R_{of}	Refer to Figs.
										At Input	At Output				
Voltage	Series-shunt	V_i	V_o	V_f	V_s	$\frac{V_o}{V_i}$	$\frac{V_f}{V_o}$	$\frac{V_o}{V_s}$	Thévenin	By short-circuiting port 2 of feedback network	By open-circuiting port 1 of feedback network	a voltage, and find the open-circuit voltage at port 1	$R_i(1 + A\beta)$	$\frac{R_o}{1 + A\beta}$	10.6 10.12 10.14 10.15
Current	Shunt-series	I_i	I_o	I_f	I_s	$\frac{I_o}{I_i}$	$\frac{I_f}{I_o}$	$\frac{I_o}{I_s}$	Norton	By open-circuiting port 2 of feedback network	By short-circuiting port 1 of feedback network	a current, and find the short-circuit current at port 1	$\frac{R_i}{1 + A\beta}$	$R_o(1 + A\beta)$	10.8(a) 10.28 10.29 10.30
Transconductance	Series-series	V_i	I_o	V_f	V_s	$\frac{I_o}{V_i}$	$\frac{V_f}{I_o}$	$\frac{I_o}{V_s}$	Thévenin	By open-circuiting port 2 of feedback network	By open-circuiting port 1 of feedback network	a current, and find the open-circuit voltage at port 1	$R_i(1 + A\beta)$	$R_o(1 + A\beta)$	10.10(a) 10.18 10.20 10.21
Transresistance	Shunt-shunt	I_i	V_o	I_f	I_s	$\frac{V_o}{I_i}$	$\frac{I_f}{V_o}$	$\frac{V_o}{I_s}$	Norton	By short-circuiting port 2 of feedback network	By short-circuiting port 1 of feedback network	a voltage, and find the short-circuit current at port 1	$\frac{R_i}{1 + A\beta}$	$\frac{R_o}{1 + A\beta}$	10.11(a) 10.24 10.25 10.26

Chapter #10: Feedback

Basic negative operational amplifiers



Assume that loop gain $A\beta$ is large.

$$A_f = 1/\beta$$

This value should serve for **final check** on A_f .

β , R_{if} , R_{of} can be determined by practical negative feedback amplifier calculation. (Figure 10.15, 10.21, 10.26, 10.30)

Thanks